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BEYOND THE ATMOSPHERE

EARLY YEARS OF SPACE SCIENCE

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Prophets and Pioneers of Spaceflight

The rocket apparently made its debut on the pages of history as a fire arrow used by the Chin Tartars in 1232 for fighting off a Mongol assault on Kai-feng-fu. The lineage to the immensely larger rockets now used as space launch vehicles is unmistakable.[1] But for centuries rockets were in the main rather small, and their use was confined principally to weaponry, the projection of lifelines in sea rescue, signaling, and fireworks displays. Not until the 20th century did a clear understanding of the principles of rockets emerge, and only then did the technology of large rockets begin to evolve. Thus, as far as spaceflight and space science are concerned, the story of rockets up to the beginning of the 20th century was largely prologue.

Nevertheless, well before the 1900s numerous authors showed a keen appreciation of what satellites might mean if only a way could be found to launch them. Their fictional accounts of space travel are often cited as early harbingers of the modern space age,[2] and in the light of recent achievements the long history of rockets makes exciting reading. Especially those engaged in space research and exploration find peculiar fascination in reading about Kepler's imaginary visit to the moon, described in the little book *Somnium, sive Astronomia Lunaris* published in 1634, several years after Kepler's death,[3] or in following the flights of fantasy recounted in Jules Verne's *De la Terre à la Lune* (1865) and *Autour de la Lune* (1870). Even the artificial satellite turned up in *The Brick Moon* by Edward Everett Hale, serialized in the *Atlantic Monthly* in 1869 and 1870. Launched by huge rotating water wheels, the Brick Moon, a manned satellite, was intended to serve as a navigational aid. In the first years of the space program, John Nicolaides, one of the engineers professionally interested in geodesy and navigation, took great delight in giving his colleagues copies of Hale's little story.

These and numerous other writings of the kind legitimately belong to the lore of the space age. While they predate the emergence of the serious work on large rockets that made the space program and space science possible, they nevertheless have a special significance. Such imaginings reflect the centuries-long interest of mankind in the heavens. Some men climbed mountains to set up astronomical observatories, others to measure how air pressure changes with height. No sooner had the Montgolfier brothers in 1783 demonstrated the feasibility of hot-air balloons than aeronauts began to fly in them. That same year J. A. C. Charles of France ascended in a hydrogen-filled balloon. This first grasp on the age-old dream of flight brought forth an amazing variety of ideas and experiments, and by the end of the 19th century powered balloon flight was a reality, the sausage-shaped dirigible being the most successful form.[4] In the 1920s and 1930s high-altitude ballooning was serious business, with men like Gray, Piccard, Anderson, and Stevens setting one altitude record after another. The record of 22 kilometers gained by the last two in the helium-inflated *Explorer II*, in 1935, persisted for two decades.

Where men could not go they sent their instruments, surrogates for the time being for those who would surely follow later. As long ago as 1749, three years before Benjamin Franklin's famous experiment with lightning, Alexander Wilson of Scotland sent thermometers aloft on kites to measure upper-air temperatures. By the late 19th century meteorologists were flying kites and balloons carrying thermometers and pressure gauges to investigate properties of the atmosphere. In 1898 the French meteorologist Léon Philippe Teisserenc de Bort started using such balloons to obtain reliable temperature measurements up to a height of 14 kilometers. By 1904 he had proved the existence of a stable, isothermal region above 11 kilometers, to which he gave the name *stratosphere*.[5]

During the 20th century free-flying balloons became a much used means of making remote scientific observations in the high atmosphere. The airplane, of course, made flying at great heights routine. While aircraft could not beat the balloons in altitude, the ability of the airplane pilot to control accurately the time and location along the flight path added a new dimension not afforded by the free balloon.

As soon as men could get to some hitherto inaccessible spot, they went there for a variety of motivations—curiosity, perhaps to make scientific observations, simply to overcome a challenge, to lay claim to a new dominion, or in pursuit of an inherent drive that it was psychologically impossible to deny. The record indicated that when men could leave the earth they would do so. The early stories about space travel served notice that man was indeed taking aim at the stars.

But, while the writings on space up through the 19th century pointed a prophetic finger toward the future, they could contribute little more. Mostly fiction, much of the writing was remarkably foresighted, but also much was incorrect. Jules Verne's enormous cannon used to launch his mooncraft from an underground pit in Florida would have subjected the passengers to bone-crushing accelerations and the spacecraft to searing temperatures from atmospheric friction. But his use of small reaction rockets

to control the attitude of the spacecraft in flight was quite correct in concept. So, too, was Achille Eyraud's concept of a reaction motor to launch a spacecraft on his *Voyage à Venus* (published in 1875), but Eyraud was not aware that water, which he used as the propellant, was utterly inadequate for the job. Likewise, Hale's water wheels could never have imparted the necessary impulse to launch the Brick Moon into orbit.

The writers of the 19th century showed awareness of the science of their day, but not enough was known about the basic physics of rockets for them to understand clearly what was required to launch spacecraft. Not until a genuine understanding of the principles of reaction motors was attained could the large rocket needed for spaceflight be created. This understanding had to await the 20th century. Those who wrote before that time were accordingly cast in the role of prophets. After them came the pioneers.

Not that prophecy ceased. Far from it. Imaginative writing continued unabated, even accelerated, as later authors found themselves able to draw upon an expanding knowledge of rockets and principles of spaceflight to give their narratives plausibility and persuasiveness. Movies, comic strips, radio programs—and later television—picked up the theme. Perhaps the culmination of these prophetic writings was to be found in Arthur C. Clarke's delightful *The Exploration of Space*—honored as a Book-of-the-Month Club selection—in which the author was able, in layman's terms yet without sacrificing technical validity, to lay before the reader a veritable blueprint of the space program to come.[6]

Much of what Clarke wrote in 1951 drew upon the pioneering work of the previous 50 years when the large rocket had been brought into being.[7] During that period mathematicians and physicists developed a sound theory of rocket propulsion. In the United States, Germany, and Russia both amateurs and professionals experimented with the design, construction, and launching of both liquid- and solid-propellant rockets. The importance of their pioneering work went largely unrecognized; the general public was mostly unaware of what the rocketeers were up to except when spectacular tests of the new fangled devices, which often ended in mishaps, caught the attention of the newsreels and newspapers. But for the seriously interested there was a growing literature to seize upon and devour.

Three persons were particularly significant in the transition from the small rockets of the 19th century to the colossi of the space age: Konstantin E. Tsiolkovsky in Russia, Robert H. Goddard in the United States, and Hermann Oberth in Germany. It is generally agreed that priority goes to Tsiolkovsky (1857-1935), who apparently in his teens became interested in the possibility of spaceflight. He wrote of spaceflight in science fiction, but went further. Self-taught in mathematics, astronomy, and physics, he proceeded to develop the basic theory of rocket propulsion, and in 1898 submitted his now famous article, "The Investigation of Outer Space by Means of Reaction Apparatus," to the editors of *Science Survey*. The article, however, was not published until 1903. For the next 30 years Tsiolkovsky continued to write both technical papers and science fiction, much of what he had to say being devoted to a favorite theme of flight into deep space, about which he wrote in 1911:

> To place one's feet on the soil of asteroids, to lift a stone from the moon with your hand, to construct moving stations in ether space, to organize inhabited rings around Earth, moon and sun, to observe Mars at the distance of several tens of miles, to descend to its satellites or even to its own surface—what could be more insane! However, only at such a time when reactive devices are applied, will a great new era begin in astronomy: the era of more intensive study of the heavens.[8]

In 1926 Tsiolkovsky suggested the use of artificial earth satellites, including manned platforms, as way stations for interplanetary flight, and in 1929 he put forth an idea for a multistage rocket which he described as a rocket train.[9]

Like the appearance of his first article on rocket principles, Tsiolkovsky's influence in Russia was delayed. As G. A. Tokaty, aerodynamicist and chief rocket scientist of the Soviet Government in Germany (1946–1947), commented:

> Konstantin Eduardovich Tsiolkovsky (1857-1935), the man of "great efforts and little rewards," is . . . considered to be the "father" of present Soviet achievements in rocket technology. He gave Russia a spaceship project which was, for 1903, absolutely unique. But being what he was—a mere teacher in a remote provincial school, a technologist rather than a theoretician—his project did not attract the attention it deserved.[10]

Apparently it took the publication in Germany, in 1923, of *Die Rakete zu den Planetenräumen*[11] by the Hungarian-born Hermann Oberth to goad the Russians into action. Following the appearance of Oberth's work, in which the author elaborated in great detail the application of rocket propulsion to spaceflight, Tsiolkovsky's earlier works were sought out and avidly studied. Interest in rocket propulsion increased noticeably in the Soviet Union, which took special pains to assert Russian claims to priority by issuing in 1924 German translations of Tsiolkovsky's writings. That same year Friedrikh A. Tsander, Tsiolkovsky, and Felix E. Dzherzhinsky started the Society for Studying Interplanetary Communications, a major aspect of which concerned interplanetary travel.

After a period of grouping and regrouping, Soviet workers in the early 1930s settled down to serious experimenting with large rockets, with which the now familiar names of F. A. Tsander, Yu. V. Kondratyuk, and M. K. Tikhonravov were associated. As early as 1928 Kondratyuk had put forth the

idea of using aerodynamic forces to slow down a rocket returning from a trip in space. Tsander designed and built a rocket motor using kerosene and liquid oxygen as propellants, which he successfully tested in 1932. In August of the following year the first successful flight of a Soviet liquid-propellant rocket took place. It was during this period that S. P. Korolev, who was to become the giant of Soviet modern rocketry in the 1940s and 1950s, began his work on rockets. His book *Rocket Flight in the Stratosphere* was published in 1934 by the USSR Ministry of Defense. But not long thereafter a curtain fell over Russian rocket activities, not to rise again until the launching of *Sputnik 1* revealed how much the Soviet Union had accomplished in the intervening 20 years.[12]

Of special interest to space scientists, during 1935 a Soviet liquid-propellant meteorological rocket designed by Tikhonravov was flown. Apparently, however, as in the United States, rocket research in the very high atmosphere had to await the availability of the more capable rockets that appeared in World War II. According to Tokaty, the exploration of the upper atmosphere with rockets of the V-2 class began in the autumn of 1947, and from 1949 on was continued with Pobeda rockets, described as greatly improved versions of the V-2.[13]

Second in priority among the rocket pioneers was the American physicist Robert H. Goddard (1882–1945), whose esteem in the United States today matches that of Tsiolkovsky in Russia. Goddard himself points to 19 October 1899 as the date when he, still in high school, determined to devote his career to the attainment of space exploration.[14] Like Tsiolkovsky and Oberth, Goddard clearly perceived the importance of rockets for high-altitude flight and astronautics. Almost immediately he began writing on the subject. Many of his papers were devoted to rocket theory, which he correctly expounded. Perhaps his most famous was the paper "A Method of Reaching Extreme Altitudes," the title reflecting his enduring interest in high-altitude and space research. The paper was originally written in the summer of 1914 and revised in late 1916 in the light of experimental results. With a few editorial changes and the addition of some notes, Goddard submitted the paper in 1919 to the Smithsonian Institution, and it was published in the *Smithsonian Miscellaneous Collections* of December 1919.[15]

Robert Goddard was set off from his contemporaries, Tsiolkovsky and Oberth, in that he by no means stuck to theory and writing as did the other two. From the start Goddard was busy with his hands, conducting experiments to check theory and devising hardware to put the theory into practice. The very year, 1914, in which he composed the first draft of the Smithsonian paper, he was awarded patents for a rocket using solid and liquid propellants, and for a multistage or step rocket. Goddard built and flew the first successful liquid-propellant rocket. Of primitive design and construction, the rocket flew 56 meters in 2½ seconds at Auburn, Massachusetts, on 16 March 1926.[16] In the course of his career he accumulated many ideas that came to be familiar features of successful large rockets—such as liquid-propellant motors, self-cooled motors, the use of gyroscopes for guidance and control, reflector vanes in the rocket jet for stabilizing and steering the rocket, fuel pumps, and parachutes for recovering a spent rocket. So prolific was his output that those who followed could hardly take a step without in some way infringing on one or more of his patents, a fact recognized by the United States government when in 1960 the military services and the National Aeronautics and Space Administration awarded $1 000 000 to the Goddard estate.

One might accordingly suppose that Goddard's influence on the space program and space science would be great, even to the extent of eclipsing that of other contributors. Unfortunately that does not appear to have been true. A suspicious nature—first aroused by adverse publicity connected with his Smithsonian paper and later reinforced by the conviction that his work was being plagiarized—led Goddard to work in isolation and for the most part to avoid open publication of his ideas and accomplishments. This secretiveness stood in the way of his contributing the leadership that he could so easily have given to the field and to the enthusiasts of the young American Rocket Society, which was founded in 1930 as the American Interplanetary Society. Years later G. Edward Pendray, one of the founders of the society, wrote plaintively: "When Goddard in his desert fastness in New Mexico proved uncommunicative, those of us who wanted to do our part in launching the space age turned to what appeared the next best source of light: the *Verein für Raumschiffahrt*—the German Interplanetary Society—in Berlin."[17]

The amateurs were not alone in their failure to join hands with the great pioneer. Members of the California Institute of Technology Rocket Research Project, established in 1936 by Theodore von Kármán, director of the institute's Guggenheim Aeronautical Laboratory, tried to persuade Goddard to join forces with them. When it was stipulated that a partnership would require mutual disclosure of ideas and projects, Goddard shied away. His reluctance to work openly with others deprived Goddard not only of the opportunity to provide leadership in the field, but also cut him off from the kind of professional assistance that he might have received from experienced engineers who could have helped put his many ideas into practice. In turning away from the Rocket Research Project, Goddard was also turning down the kind of funding support from the military that could have capped his long years of work with their hoped-for fruition. Working alone with extremely limited funds, Goddard could not match the progress being made in German rocketry, which was supported amply by the military during those years.

Goddard furnishes a tragic illustration of the importance of open publication and free exchange of ideas to the scientific process. Unpublished, the import of Goddard's ideas went unrecognized for the most part, and by the time they were widely known much of what he had done had been redone, as with the V-2. The opportunity to be the leader of the field during the course of his development work soon passed. Still, he was a man of genius and originality and the many honors later accorded him were well deserved. NASA's Goddard Space Flight Center in Greenbelt, Maryland, appropriately bears his name. Medals are awarded and symposia held in his honor. In 1958 the National Rocket Club began sponsoring the Robert H. Goddard Annual Memorial Dinner in Washington, faithfully attended by engineers, scientists, administrators, legislators, military men, industrialists—the Who's Who of rocket and space research—to pay tribute to Goddard's pioneering role. Space scientists also recognize in Goddard the first to work seriously on the problem of developing an effective means of sending scientific instruments beyond balloon altitudes into the upper atmosphere and outer space.

But Goddard never did personally achieve his dream of using rockets for upper-atmosphere research. While he continued to work in obscurity, spending his final years during World War II working in secrecy for the U.S. Navy, the CalTech Rocket Research Project—reorganized in 1944 as the Jet Propulsion Laboratory—went on to become the first group in the United States to build and launch a rocket specifically designed for upper-air research. Named the WAC-Corporal, the JPL rocket on 26 September 1945 rose to a height of about 70 kilometers, a U.S. record at the time. It is the JPL research, rather than Goddard's, from which a line can be traced directly to the space program. Writers associated with CalTech and the von Kármán group communicated the latest in rocketry to the public through scientific papers.[18] Although restricted in its circulation at the time, because of its bearing on military applications, a handbook of jet propulsion put out by JPL nevertheless reached large numbers of persons in rocket research and development.[19] More significantly for space science, the WAC-Corporal was the progenitor of a larger, improved sounding rocket, called Aerobee—in later versions capable of carrying a substantial instrument load above 200 kilometers—which became one of the mainstays of the American high-altitude research program.[20]

Neither Goddard's work nor the JPL rockets provided the initial impetus to the space science program in America. Circumstances made rocket sounding in the United States the beneficiary of the two decades of vigorous rocket development work by German experimenters that ensued following the publication of Oberth's *Rocket into Planetary Space*. Nourished by German military support, the German experimenters rediscovered and reinvented for themselves much of what Goddard was learning in the United States. Going well beyond what Goddard could accomplish in his self-imposed isolation, Walter Dornberger, Wernher von Braun, and their colleagues produced the V-2—Vergeltungswaffe-Zwei or "Vengeance Weapon

Two"—the first large rocket to see substantial service.[21] At the close of World War II, U.S. Army forces captured large numbers of these monsters at underground factories in the Harz Mountains in central Germany. Along with von Braun and key members of his team—who took the initiative to ensure that they became prisoners of American, not Russian, forces[22]—the Army took the captured V-2s to the United States. There the missiles were assembled, tested, and launched at the White Sands Proving Ground in New Mexico to provide experience in the handling and operation of large rockets.

Rather than fire the missiles empty, the Army offered to allow interested groups to instrument them for high-altitude scientific research. A number of military and university groups accepted, forming the V-2 Upper Atmosphere Research Panel, which became the aegis for the country's first sounding rocket program.[23]

4

The Rocket and Satellite Research Panel: The First Space Scientists

As World War II came to a close, a group of engineers and scientists in the Communications Security Section of the Naval Research Laboratory in Washington began to cast about for new research problems to which to apply their talents. Long hours were spent on the subject, and the list of possibilities grew to sizable proportions. Milton Rosen, a competent, versatile, imaginative electronics engineer, suggested that the group might apply its wartime experience with missiles and communications, including television, to a study of the upper atmosphere. The suggestion became the eighth to go on the blackboard in the office of Ernst Krause, head of the section. Thereafter it was referred to as Project 8.

When the debate finally wound down, Project 8 was the clear winner. To the many physicists in the group the project offered an attractive and important field of research. The engineers could feel the challenge of instrumenting and launching the rockets that would be needed by the scientists. And because of the importance of knowledge of atmospheric properties to communications and the design and operation of missiles, it was possible that the Navy might support the project.

The director of the laboratory approved the upper-air research proposal in December of 1945, and the section became the Rocket Sonde Research Section, a name that appropriately enough also came from the originator of the Project 8 idea. No one in the section was experienced in upper atmospheric research, so the section immediately entered a period of intensive self-education. Members lectured each other on aerodynamics, rocket propulsion, telemetering—whatever appeared to be important for the new tasks ahead. The author gave a number of talks on satellites and satellite orbits. Indeed, the possibility of going immediately to artificial satellites of the earth as research platforms was considered by the group, which assimilated carefully whatever information it could obtain from military studies of the time. The conclusion was that one could indeed begin an artificial satellite program and expect to succeed, but that the amount of new development required would be costly and time consuming. The scientists could not hope to have their instruments aloft for some years to come and, anyway, were not likely to get their hands on the necessary funds. The Rocket Sonde Research Section accordingly shelved the satellite idea and turned to sounding rockets.

As they were considering what rockets—including the Jet Propulsion Laboratory's WAC-Corporal—might be available for the research they contemplated, word came that the U.S. Army would be willing for interested scientists to conduct experiments in some of the V-2s it was planning to fire at the White Sands range in New Mexico. Because of the narrow confines of the range, the missiles would have to be fired along nearly vertical trajectories and would accordingly make ideal probes of the upper atmosphere. To explore the possibilities Krause invited a number of interested persons to meet at the Naval Research Laboratory. At the meeting, on 16 January 1946, physicists and astronomers interested in cosmic ray, solar, and atmospheric research were present. Because of the potential importance of upper-air data to military applications, the services were well represented. It was plain from the deliberations that a number of groups both in universities and in the military would be interested in taking part in a program of high-altitude rocket research.

The V-2 Panel

Accordingly, at an organizing meeting at Princeton University 27 February 1946, a panel was formed of members to be actually engaged in or in some way directly concerned with high-altitude rocket research.[1] The original members (see also app. A) were:

E. H. Krause (chairman), Naval Research Laboratory
G. K. Megerian (secretary), General Electric Co.
W. G. Dow, University of Michigan
M. J. E. Golay, U.S. Army Signal Corps
C. F. Green, General Electric Co.
K. H. Kingdon, General Electric Co.
M. H. Nichols, Princeton University
J. A. Van Allen, Applied Physics Laboratory, Johns Hopkins University
F. L. Whipple, Harvard University

Because of his role in getting things started and because he would be devoting full time to upper-air research with rockets, Krause was elected chairman.

To Krause must go the principal credit for getting the program under way. He was a physicist, with a doctorate from the University of Wisconsin in spectroscopy, and a background in communications research. Both qualifications were pertinent to the development of techniques for the investigation of the sun and upper atmosphere. Krause's energy and drive were phenomenal, and his capacity for detail and thoroughness were ideally

suited to welding all the elements needed to get a sounding rocket program off the ground. When Krause left in December 1947 to participate in nuclear bomb tests, James A. Van Allen was elected to the chair, a spot he occupied for the next decade.[2]

Van Allen is by far the best known of the original members of the V-2 panel. A physicist, at the time the panel was formed he was employed by the Applied Physics Laboratory of the Johns Hopkins University on the Bumblebee Project, a Navy missile research and development project. He brought to the panel an intense interest in cosmic ray physics, an interest that led in time to his discovery of the earth's radiation belts that now bear his name.

The panel had no formal charter, no specified terms of reference from an authorizing parent organization, a circumstance that left the panel free in the years ahead to pursue its destiny in keeping with its own judgment. The immediate task was to provide Col. James G. Bain of the Army Ordnance Department with advice he had requested on the allocation of V-2s to the various research groups. This the panel proceeded at once to do, and in fact until the end of the V-2 program in 1952 continued to direct its reports to Army Ordnance as principal addressee. Thereafter the reports were issued simply to the members and to observers who attended the meetings, with copies to a selected list of interested persons and agencies (see app. B).

The panel's program, if it may be called that, consisted of the collection of activities engaged in by its members. As a forum for discussion of past results and future plans, the panel was a breeding ground for ideas; but whatever control it might bring to bear on the program was exerted purely through the scientific process of open discussion and mutual criticism.

For some time after its first session, the panel met monthly (see app. C). There was a great deal to do, quickly; for Army Ordnance and its contractor, General Electric Company, intended to fire the rockets on a rather rapid schedule. Since the German warheads were not suitable for carrying scientific payloads, the Naval Research Laboratory undertook to provide the different groups with standard nose sections specifically designed for housing the research instrumentation. To send information to the ground from the flying rocket, NRL also furnished telemetering equipment to go into the rocket and erected ground stations at the White Sands range for receiving and recording the data-bearing signals. In short order the word *telemetering*, meaning the making of remote measurements by radio techniques, became a familiar part of the growing jargon of rocket sounding. To make the most of the large capacity of the V-2, NRL designed and built a large, complex telemeter. The first version supplied to the program could provide 23 channels of information; a later version provided 30. With characteristic preference for smaller, simpler instrumentation, the Applied Physics Laboratory developed and used a much smaller, 6-channel, frequency-modulated telemeter.[3]

Radar beacons were installed in the missile to track it, providing information on where measurements had been made. The range also required that each rocket be outfitted with a special radio receiver that could cut off the motor should the missile begin to misbehave after launch. Arrangements had to be made for building and supplying this equipment. Also, to supplement the tracking information provided by radar and radio, theodolites, precise cameras, and other optical instruments were installed at strategic locations along the firing range to furnish both visual and photographic trajectory data. It also would be essential to know the orientation of the rocket in order to interpret properly such measurements as aerodynamic pressures or cosmic ray fluxes. For this, still more instruments—including photocells to observe the direction of the sun, cameras, and magnetometers—were brought to bear.[4]

Although much, perhaps most, of the scientific data would be obtained by telemetering, some measurements would require the recovery of equipment and records from the rocket after the flight was over, such as earth and cloud pictures, photographs of the sun's spectrum, and biological specimens exposed to the flight environment. For this purpose several techniques were developed, including the use of explosives to destroy the streamlining of the rocket, causing it to maple leaf to the ground; the deployment of parachutes to recover part or all of the spent rocket; and even the application of the kind of sound ranging techniques used in World War I to locate large guns.[5]

At first, operations at White Sands were an amorphous collection of activities. During the first year of rocket sounding the procedures and issues that would have to be dealt with in even greater detail years later in the space program emerged: safety considerations, provision for terminating propulsion of the missile in mid-flight, tracking, telemetering, timing signals, range communications, radio-frequency interference problems, weather reports, recovery of instruments and records, and all that went into assembling, instrumenting, testing, fueling, and launching the rocket. To cope with the seemingly endless detail, the range required formal written operational plans in advance that could be disseminated to the various groups. A more or less standard routine evolved with which the participants became familiar.[6] In only a few years experimenters were harking back to the "good old days" when operations were free and easy and red tape had not yet tied everything into neat little, inviolable packages.

While the General Electric Company personnel, Army workers, and others labored to produce successful rocket firings, the scientists labored equally hard to devise and produce the instrumentation that would yield the desired scientific measurements. At first some of the instrumentation was tentative, even crude, as when Ralph Havens of NRL took an auto-

mobile headlight bulb, knocked off the tip, and used it as a Pirani pressure gauge to measure atmospheric pressure in the V-2 fired on 28 June 1946. But even before the end of 1946 spectrographs were recording the sun's spectrum in previously unobserved ultraviolet wavelengths, special radio transmitters were measuring the electrification of the ionosphere, and a variety of cosmic-ray-counter telescopes were analyzing radiation at the edge of space. A portion of each panel meeting was devoted to reporting on experimental results, which accumulated steadily from the very first flight of 16 April 1946. Papers began to appear in the literature and attracted considerable attention as experimenters reported on measurements that hitherto were impossible to make.[7] By the time the last V-2 was fired in the fall of 1952, a rich harvest of information on atmospheric temperatures, pressures, densities, composition, ionization, and winds, atmospheric and solar radiations, the earth's magnetic field at high altitudes, and cosmic rays had been reaped.[8]

THE NEED TO REPLACE THE V-2

But not all of the results had been obtained from the V-2. To be sure, the immediate availability of the V-2 as a sounding rocket was a boon to the program, for it meant that the scientists could start experimenting without delay. Its altitude performance of 160 kilometers with a metric ton of payload far exceeded that of any other rocket that the experimenters might have been able to use, making investigations well into the ionosphere possible from the outset. More significantly, the large weight-carrying capacity of the rocket meant that experimenters did not have to miniaturize and trim their equipment to shoehorn them into a very restricted payload, but could use relatively gross designs and construction. This capacity was a great help at the start, when everyone was learning, for it permitted the researcher to concentrate on the physics of his experiment without being distracted by added engineering requirements imposed by the rocket tool. Later, with some years of experience behind him, the experimenter would be able to take the outfitting of much smaller rockets in stride. And it was of advantage to go to smaller rockets as soon as possible.

Smaller rockets would be much cheaper, far simpler than the V-2 to assemble, test, and launch. Moreover, with the smaller, simpler rockets the logistics of conducting rocket soundings at places other than White Sands would be manageable. With such thoughts in mind, as panel members pressed the exploration of the upper atmosphere with the V-2 they also set out to develop a variety of single and multistage rockets specifically for atmospheric sounding.[9] James Van Allen and his colleagues at the Applied Physics Laboratory undertook, with support from the U.S. Navy's Bureau of Ordnance, to develop the Aerobee sounding rocket.[10] At the same time NRL took on the job of developing a large rocket—first called Neptune, but later Viking when it was learned a Neptune aircraft already existed—to replace the V-2s when they were gone.[11] At the 28 January 1948 meeting of the panel, Van Allen reported on a series of test firings of the Aerobee—three dummy rounds and one live round.[12] As soon as it was ready the Aerobee was put to work exploring the upper atmosphere and space, with firings not only from the original Aerobee launching tower at White Sands, but also from a second tower that the Air Force erected some 57 kilometers northeast of the Army blockhouse at the White Sands Proving Ground. The Air Force tower was located at Holloman Air Force Base near Alamogordo. Not content with the payload and altitude capabilities of the first Aerobees, both the Air Force and the Navy continued the development, producing something like a dozen different versions, one of which could carry 23 kilograms of payload to an altitude of 480 kilometers.[13] In its various versions Aerobee was used continuously in the high-altitude rocket research program through the 1950s and 1960s and was still in use in the mid-1970s.

In contrast, the Viking, although of a marvelous design—Milton Rosen, who directed the Viking development program, used to point out that in its time Viking was the most efficiently designed rocket in existence—found very little use. The dozen rockets bought for the development program were, of course, instrumented for high-altitude research. But Viking was too expensive. The groups engaged in rocket sounding each had perhaps a few hundred thousand dollars a year to expend on the research, and a single Viking would have eaten up the whole budget. When the supply of German V-2s began to run low, consideration was given to building new ones; but estimates placed the price per copy at around half a million dollars, which was prohibitive. It had been hoped that Viking would be much less expensive, but before the end of the development these rockets became almost as expensive as new V-2s. So Viking found no takers among the atmospheric sounding groups and would probably have been shelved had it not been chosen as the starting point for the Vanguard IGY satellite launching vehicle.[14]

The contrast between Viking and Aerobee typified a situation that has recurred in the space science program. One group of scientists would favor developing large new rockets, spacecraft, or other equipment that would greatly extend the research capability. Another group would prefer to keep things as small and simple as possible, devoting its funds to scientific experiments that could be done with available rockets and equipment. The former group could always point to research not possible with existing tools, thus justifying the proposed development. In rebuttal the latter could always point to an ample collection of important problems that could be attacked with existing means. There was right on both sides of the argument, and it was usually a standoff. As far as upper atmospheric research was concerned, however, Viking was too far ahead of its time.

While in the next decade researchers would be able to buy $1-million Scouts (chap. 10), in the early years of rocket sounding Viking cost too much.

Once the ball had started rolling with Aerobee and Viking, other rocket combinations began to appear. The experimenters sought less cost, greater simplicity, higher altitudes, more payload, and especially a capability to conduct firings at different geographic locations. Great ingenuity was displayed in putting together new combinations. Sounding rockets were taken to the California coast, to Florida, to the Virginia coast, out to sea, and to the shores of Hudson's Bay in Canada.[15] They were even launched in the stratosphere from balloons, a combination that the inventor, Van Allen, called a Rockoon.[16] In the panel meeting of 9 September 1954, Van Allen reported that Rockoon flights in the Arctic had established the existence of a soft radiation in the auroral zone above 50 kilometers height, which proved to be one of the milestones along the investigative track that ultimately led to the discovery of the earth's radiation belt.

SCOPE OF PANEL ACTIVITY

One of the most notable aspects of the panel record is the steadily increasing scope of activity. In the minutes of the organizing meeting, the secretary referred to the group simply as "the panel." By the third meeting Megerian was calling the group the "V-2 Upper Atmosphere Panel." This name continued for the next two meetings; but the appellation "V-2 Upper Atmosphere Research Panel" appeared at the sixth meeting, in September 1946, and stuck for the next year and a half. These first titles reflected the panel's participation in the V-2 program, but the group's primary business was high-altitude research, not V-2s. The panel, well aware that the supply of V-2s would be exhausted in the not too distant future, gave early attention to finding alternative sounding rockets. Prodded by the Office of the Chief of Ordnance, at its March 1948 meeting the panel dropped the V-2 from its title and began calling itself the "Upper Atmosphere Rocket Research Panel" (UARRP). This sufficed to describe activities until members had become so thoroughly involved in the International Geophysical Year scientific satellite program that another name change seemed appropriate. At an executive session, 29 April 1957, the panel adopted its final name: "Rocket and Satellite Research Panel."[17]

Throughout most of its active life, the panel remained quite small. By restricting its rolls to working members only, and also by limiting the number of representatives from any one agency, the panel kept its size down—which made for more manageable meetings. Yet there was no desire to limit interest or participation in the meetings. A loyal cadre of observers attended the sessions throughout the years and joined in the discussions. From the first, the National Advisory Committee for Aeronautics was represented among the observers—an interesting fact in retrospect, although at the time there was no reason to suspect that one day NACA might play a central role in a suddenly emerging space program. Increasing interest in high-altitude rocket research over the years is also shown by the steady growth in the list of addressees to whom panel reports were sent. The minutes of the organizing meeting went to only about 30 persons; 10 years later some 118 copies were being distributed among 73 addressees.[18] The composition of the distribution lists is illuminating (see app. B). The military was obviously interested. So, too, were other government agencies such as NACA and the U.S. Weather Bureau. The large number of university names on the list no doubt resulted from the pure-science nature of much of the panel's research.

For more than a decade the panel occupied a unique position in scientific research. In the United States its members represented all the institutions engaged in sounding rocket research. Attendees at meetings—members plus observers—comprised a substantial number of the individuals in the country who were involved. As one consequence of this unique position, the panel came to be regarded as the prime source of expertise in the field. In spite of the lack of any official charter, the panel soon acquired a quasi-official status. The National Advisory Committee for Aeronautics used data from the panel program in compiling and updating its tables of a standard atmosphere.[19] The Defense Department's Research and Development Board made a practice of turning to the panel for recommendations regarding sounding rockets and high-altitude rocket research. The board—called the Joint Research and Development Board before the establishment of the Department of Defense in 1947—boasted a sprawling, complex structure intended to correspond in one way or another to the military research and development programs.[20] From time to time its Committee on Guided Missiles took an interest in the rockets being used by the panel. When, in the spring of 1949, the Navy's Viking and the Air Force's MX774 rockets came into competition—it was not considered reasonable for the country to support two large, expensive sounding-rockets—UARRP was informed that a panel of the Committee on Guided Missiles endorsed Viking. The R&D board's Committee on Geophysical Sciences, and its subsidiary group for study of the upper atmosphere, took a continuing interest in what UARRP was up to. The subsidiary group endorsed the UARRP's research program and in November 1947, responding to a request for support, unanimously recognized "the importance of all phases of the well-coordinated V-2 rocket firings program and the grave consequences of any failure to give adequate financial support to all agencies involved in this program, since the lack of support of the program in any one agency would jeopardize the program as a whole."[21] At its April 1950 meeting, one finds the UARRP responding

to a request of the R&D board for views on requirements for upper-air research vehicles.[22]

But, while the endorsement was of help, association with the military also brought problems. At its 7 May 1947 meeting, the UARRP learned that the R&D board's upper-atmosphere group was considering assigning primary responsibility to different agencies for different kinds of upper-atmosphere research. Although nothing ever came of this, the thought of dividing the research into assigned parcels conflicted with the basic research instincts of UARRP members.

More serious, however, was the question of security classification that arose periodically. In defense of the research program, panel members were accustomed to pointing out the many practical benefits to be gained from data and knowledge obtained. Over the years the list of potential benefits to the military grew, until a report issued at the start of the International Geophysical Year by a number of the panel members cited a dozen important applications:

- Design of missiles, high-altitude craft, and space vehicles.
- Determination of the reentry behavior of long-range ballistic missiles.
- Special techniques of high-altitude navigation.
- Evaluation of hazards to personnel and equipment in the high atmosphere and space.
- Improvement of weather forecasting.
- Study of climate.
- Prediction of the trajectories of biological, chemical, or radiological agents.
- Development of reliable point-to-point communications.
- Development of reliable and accurate methods of guidance, control, and delivery of missiles to their targets.
- Development of reliable and accurate methods of detection of enemy missiles and high-altitude craft.
- Development of countermeasures against enemy missiles.
- Remote detection of nuclear explosions.[23]

But to the extent the salesmanship succeeded, it also raised the question of why the sounding rocket results shouldn't be classified if they were so valuable to the military, which was paying for them.

From the outset the panel had assumed that its program, being basic research, would be unclassified. In a memorandum to the White Sands

Proving Ground, Col. H. N. Toftoy of the Army Ordnance Department had written that V-2 firing schedules, rocket design, and flight information would be unclassified.[24] This decision was important to the program, since the flight information was intimately related to the high-altitude data obtained from the rocket, and since design data were needed for interpreting measurements—for example, aerodynamic pressure curves were required in obtaining atmospheric densities from pressure measurements along the surface of the flying rocket. A serious threat arose when, at the October 1952 meeting of the panel, Earl Droessler of the R&D board announced that the military had again raised the question of classification of upper atmospheric data. The panel unanimously agreed to fight classification, citing the importance of the scientific process, in particular open publication and free exchange of information, to a basic research activity. While there was something to be gained by classifying certain specific uses of scientific information, there was much to be lost by classifying the purely scientific data. In these efforts the panel was successful, and the program remained unclassified.

The program called for a lot of work, but it was exciting. Panel meetings were enjoyable, with none of the tedium that so often weighs oppressively on committee meetings. For most of the members, after a period of preparation at home base—in Washington, Silver Spring, Cambridge, Ann Arbor, or elsewhere—there would be a period of some weeks or a couple of months working in the lonely beauty of the New Mexico desert. How exhilarating it was to send a rocket roaring into the clear blue sky, watch the missile trace a brilliant white vapor trail against the azure background, a trail the stratospheric winds soon blew into complicated twists and knots, and then to jump into a jeep and race northward to retrieve cameras and instruments! On one such day in March 1957, with the sky as bright a blue as it ever had been, V-2 no. 21 landed in the heart of the White Sands National Monument. What a glorious hunt riding up and down over the snow-white dunes of gypsum sand that stretched as far as the eye could see! At the end of the day, with a solar spectrograph, cameras, and other instruments safely stowed aboard the jeeps, the impact party, as it was called, slowly worked its way out of the barren wilderness. As the group approached the edge of the monument, where the gypsum deposit has acquired a pinkish tint from the surrounding red sands of the Tula Rosa Basin, the sun was setting. An occasional yucca growing amid the pink-white dunes provided a display of incomparable beauty, which the glowing sun transformed into a fairyland. When the white sands were finally left behind, one could feel the emotional release.

The routine was frequently broken by bits of humor. Early in the program, before the range was properly instrumented for tracking the V-2s, von Braun often watched the flying rocket as it rose above the desert, judging by eye whether it was on course. If the missile strayed, von Braun

called for stopping the engines by radio. On one occasion, the eye failed to detect a tipping toward the south, and the missile landed in a cemetery in Juarez, Mexico, causing something of an international incident. Rumor had it that von Braun's lapse might have been related to his having some instruments riding on the rocket. At any rate preparations to track the missiles by instrument were accelerated.

The Naval Research Laboratory used radio signals from the flying rocket to measure the electrification of the ionosphere. For this purpose the laboratory installed ground stations uprange from the launching area. One day as the men were preparing one of the stations for an approaching flight, an Army jeep drove up, and a soldier got out and began driving a stake into the ground not more than a stone's throw from the station. Curious, the men asked what that meant. That, they were told, was the aiming point for some planned Honest John rocket tests. The men let it be known they didn't fully appreciate being made the target of rocket firings. "Not to worry," was the answer, "we never hit the target, anyway!"

Often there was frustration to struggle with. During the countdown for the firing of V-2 no. 16, something in the tail switch, which was supposed to turn the experimental equipment on after takeoff, was wrong. An effort was made to reconnect the switch there on the launch stand with the fully loaded rocket waiting to take off. After launch, however, instead of turning instruments on, the rewired switch proceeded to turn everything off. A postflight review showed that there were several ways in which the switch could have been connected to do the intended job, and only one way in which it would fail. The one and only wrong way had been chosen—an important object lesson regarding hasty, last-minute changes in the field. It turned out, however, that this rocket tumbled end over end in flight, which would have made the reduction of data an exceedingly complex matter. The scientist in charge later said it was probably a good thing that the equipment had been turned off, for otherwise the experimenters would surely have been unable to resist the temptation to try to interpret the measurements and probably would have wasted a lot of time on a futile exercise.

On another occasion, as a physicist watched a rocket carry aloft the cloud chamber over which he had labored long and hard, he remembered that he had forgotten to remove the lens cap from the recording camera. To add to the feeling of despair, the telemetering record indicated that the cloud chamber had worked perfectly during the flight.

Of course, it was always heartbreaking when the rocket failed to perform. It was difficult enough for some experimenters to reconcile themselves to the thought that the equipment they had struggled to perfect would often be destroyed on a single flight. There was consolation when the flight produced the data sought, but not when the rocket failed. After the program had been under way for some time, it was noted that the rockets bearing the simplest payloads seemed to have the best success. The Applied Physics Laboratory group, which never attempted to load rockets to full capacity, had acquired an image of almost perfect success. In contrast, the Air Force Cambridge Research Center, which tried to conduct dozens of complicated experiments on a single flight—and even lengthened the V-2 by a whole diameter to make additional instrument space—had developed an image of almost complete failure. The Naval Research Laboratory, which flew payloads intermediate between those of APL and AFCRL in complexity, succeeded about two-thirds of the time. There seemed to be an interaction between the experimenting and the launching operations, the more complex experiments tending to induce more problems with the rocket itself. The suspicion that this was actually happening was widely held, but never proved. On closer look, the evidence is not as clear as it seemed at the time, for the Princeton experiments were as simple as any, and yet all their rockets failed, which was no doubt the main reason for Princeton's early withdrawal from the program.

One cannot work with rockets without a certain amount of danger. Although the missiles were aimed away from them, the stations uprange were nevertheless exposed to some risk that the rocket might land on one of them. No direct hit ever did occur, but on a few occasions the wreckage from a falling rocket landed uncomfortably close. The greatest danger existed when the rocket was being loaded with propellants and people were still working around it, completing last minute preparations. When a spurt of hydrogen peroxide set a jeep afire, the industrial supplier was moved to assert publicly that the liquid was perfectly safe if only it were handled properly. Most distressing were accidents to personnel, as when a fuming sulfuric acid mixture being loaded into a V-2 prematurely ejected, spraying the face of a worker and endangering his eyesight. The acid mixture was used to generate visible clouds in the stratosphere, which were then tracked to measure stratospheric winds.

The author vividly remembers working with a companion on a platform 10 to 15 meters above ground, inserting live JATO* rockets into receptacles in the midsection of a fully loaded V-2. Tests had shown that JATO would ignite from the slightest applied voltage, and care had to be exercised not to generate any static electricity or to permit current to flow through the JATO igniter from the ohmmeter being used to check the circuits. Other workers had retired to a respectful distance. Slanting cables had been drawn between the work platform and the ground, down which—if things went wrong—one could slide and then run like hell to safety. The JATOs, which were intended to impart a spin to the rocket in the upper atmosphere, did not ignite during the loading. But, then, neither did they spin the V-2 in flight.

*Jet Assist Take Off rockets permitted heavily loaded aircraft to take off from short runways.

INTERNATIONAL CONTACTS

From the panel's labors gradually accumulated an array of answers to important questions that had previously been intractable. As noted earlier, published results began to attract attention in the United States. The sounding rocket program also aroused interest abroad. At the panel's 13 June 1950 meeting, Sydney Chapman, renowned geomagnetician from the United Kingdom, joined the discussions. From that time international contacts gradually broadened, as Chapman became a frequent participant and visitors from Belgium, Australia, Japan, and Canada came. In the fall of 1952 the Royal Society's Gassiot Committee—a committee concerned with upper atmospheric research—proposed an international meeting on that subject, to be held at Oxford the following August. At the conference the Europeans heard the U.S. program and results discussed in detail, while the Americans became aware of a growing interest among scientists from other countries. By publishing the proceedings in book form, the British stole something of a march, giving panel members occasion to reassess their own publication program.[25]

At this very period early plans for a "Third Polar Year"—a worldwide cooperative program of geophysical investigations—were taking shape (chap. 5). Van Allen and other panel members had already been considering the possibility of conducting rocket soundings in the vicinity of Fort Churchill, Canada. The author proposed at the panel's January 1953 meeting that a "full fledged operation of Northern latitude firings be organized for the Third Polar Year 1957-1958" and presented objectives and requirements for such a program at the following meeting.[26] In October 1953, Joseph Kaplan, chairman of the U.S. National Committee for the International Geophysical Year (the new name for the Third Polar Year), and Sydney Chapman, chairman of the International Committee for the IGY, both approved the idea of an IGY rocket program. Kaplan reported that the panel would be asked to serve as advisory committee to the National Academy of Sciences' National Research Council for the rocket phases of the IGY program, but very shortly thereafter the academy established its own Technical Panel on Rocketry.[27] To coordinate planning and preparations for firings at Fort Churchill—after some negotiations Canada formally extended an invitation to the United States to set up a rocket-launching range there—the panel formed a Special Committee for the IGY (SCIGY). Hearing of the Research Council's Technical Panel on Rocketry, the panel transferred SCIGY to the academy's technical group.[28] SCIGY's membership was then expanded slightly to the following:

H. E. Newell, Naval Research Laboratory, Chairman
J. W. Townsend, Jr., Naval Research Laboratory, Executive Secretary
John Hanessian, Jr., National Academy of Sciences, Recording Secretary

K. A. Anderson, State University of Iowa
Warren Berning, Ballistic Research Laboratories
L. M. Jones, University of Michigan
R. M. Slavin, Air Force Cambridge Research Center
N. W. Spencer, University of Michigan
W. G. Stroud, Signal Engineering Laboratories[29]

The military services permitted their employees to take part in the IGY program and undertook to provide logistic support for shipboard operations and for setting up an Aerobee tower and a Nike-Cajun launcher at Fort Churchill. The Army was in overall charge of the U.S. rocket contingent at Fort Churchill, while the three services shared the expenses. But additional funds were needed. Accordingly, Van Allen submitted on behalf of panel members a budget request to the IGY committee for more than one and a half million dollars, about 15 percent of America's total planned budget for the IGY.[30] The costs of the program were defrayed by both the military services and the IGY budget.

The IGY Satellite Program

Although individual members had long been interested in the use of artificial satellites for scientific research, the panel up to this point had recommended only a sounding rocket program for IGY. But simultaneously with the planning for rocket firings, enthusiastic advocates were pressing for the launching of scientific satellites. Inevitably the panel was caught up in these proposals. To explore at length the usefulness of satellites for scientific research, the panel sponsored a symposium at the University of Michigan 26-27 January 1956. The proceedings were published in a book,[31] the sale of which generated a small treasury for the panel.*

Once aroused, interest in scientific satellites grew rapidly. Most members took part one way or another in the IGY satellite program. Gradually the idea emerged that the United States should go further and establish some kind of permanent space agency. In the summer of 1957, the author jotted down some brief notes outlining a "National Space Establishment" to be organized and funded to conduct unmanned space research and applications and manned exploration of outer space. Shortly thereafter the panel—which the preceding April had changed its name to Rocket and Satellite Research Panel—took steps to explore formally its potential interest in earth satellites and outer space. Report 47, 19-20 September 1957, records the creation of a Committee on the Occupation of Space, chaired by the author. When *Sputnik 1* went into orbit, the panel intensified its efforts on behalf of a civilian National Space Establishment.[32]

On 21 November the group issued a paper entitled "A National Mission to Explore Outer Space." A different version, "National Space Establishment," appeared on 27 December 1957 (see app. D). The minutes of the 6 December panel meeting record that the earlier paper had been discussed with Detlev Bronk, president of the National Academy of Sciences. Copies had also been given to James Killian, the president's science adviser, and to Lee DuBridge, president of the California Institute of Technology. Dr. Killian referred the panel report to Herbert York, Emanuel Piore, and George Kistiakowsky, members of the President's Science Advisory Committee who were also exploring the question of the United States role in space.[33]

To this point the panel's policy of restricting membership to those working in the upper-atmosphere program had made good sense. But now the panel felt the need for additional weight behind its recommendations. During December 1957 the membership about doubled, adding key persons in the military research establishment, industry, the rocket development field, and the American Rocket Society (app. A). The society was also agitating at the time for the creation of a civilian space agency.[34] The two groups agreed to join forces in promoting the idea, and on 4 January 1958 issued a summary paper supporting their joint proposal for a "National Space Establishment" to have responsibility for investigating and exploring space.

In addition to preparing that paper, the Rocket and Satellite Research Panel mapped out a plan to bring its recommendations to the attention of persons who might be in a position to do something. Members visited congressmen and officials in the administration and sought help from the Academy of Sciences. A small group, chaired by the author and including Wernher von Braun and William Pickering, called on Vice President Nixon, who seemed most receptive. Through his good offices a number of meetings were arranged for the group on Capitol Hill and in the executive branch: with the commissioners and general manager of the Atomic Energy Commission, with George Allen and key figures in the U.S. Information Agency, and with the staffs of the House and Senate committees that were considering how to respond to the Soviet challenge in space. William Stroud, von Braun, and the author appeared before the Joint Committee on Atomic Energy and shocked members by asserting that the proposed space program could very likely require as much as a billion dollars a year and could become comparable to the atomic energy program once it got going.[35]

Panel members, of course, seized on whatever news they could acquire about what was going on. They heard that the space program could go a number of ways: a new agency might be created, which the panel had naively recommended; or responsibility might be assigned to an existing

*Having a bank account was a source of some perplexity not resolved until years later, when the money was donated to a small, nonprofit activity called Science Services. The income from the gift was to provide for an annual award to a student competing in the International Science Fair. The panel suggested that the award be for excellence in the field of space exploration, space science, space engineering, or space application. Megerian, minutes of panel, rpt. 1968-1.

agency like the National Advisory Committee for Aeronautics; or the Department of Defense might get the job.[36]

Among panel members the NACA had an image of gross conservatism. In talking to Hugh Dryden, director of NACA, Whipple received the impression that NACA was "not prepared to undertake space research on the scale considered essential by the RSRP and by the American Rocket Society." Whipple had also talked with General Doolittle, NACA chairman, who declared his intense feeling that it would be a great error to set up any such organization outside of the Defense Department's jurisdiction.[37] His opinion was disturbing to panel members, who had felt the pinch of budgets for sounding rocket research competing with budgets for purely military purposes and who would like to remove the periodic vexation of the classification battle. Although members recognized that the new agency would have to depend on the military for a great deal of hardware and logistical support, to a man—including those employed by the services—the panel was determined that the nation's space agency ought to be civilian.

Doubts about the NACA did not lessen the feeling of satisfaction with the National Aeronautics and Space Act of 1958. Members were prepared to give whole-hearted support to the new National Aeronautics and Space Administration, which was to absorb NACA as its nucleus. Indeed, many joined the new agency. But the panel itself was now at loose ends. The purposes it had served for more than a decade would now be NASA's. For the next two years the panel devoted itself to colloquia on topics related to atmospheric and space research, but such colloquia could hardly serve the now explosively expanding field the way sessions of the scientific societies could. Members experienced a growing dissatisfaction where before a feeling of pioneering excitement had suffused the discussions. William Pickering submitted his resignation with a statement that he felt that the panel no longer served any real purpose.[38]

Having existed for so long without any formal charter, the panel now found time to compose a constitution, which was declared adopted by a three-fourths vote at the meeting of 17 February 1960.[39] After one more meeting, the panel suspended operations.

Thus, the panel's success in helping bring about the creation of a new agency devoted to the investigation and exploration of space also brought the demise of the panel. In contrast, the National Academy of Sciences, which the Rocket and Satellite Research Panel had drawn into the rocket research field, expanded its role in the program after the creation of NASA. The Space Science Board, which grew out of the academy's IGY panels on rocketry and earth satellites, was an immediate source of advice to NASA in its formative years, taking over the advisory role that the Rocket and Satellite Research Panel had once played. As a committee of the nation's prestigious Academy of Sciences, the Space Science Board enjoyed a vantage point that the panel never had commanded. How the academy went into space science and events leading to the establishment of the Space Science Board as one of the prime sources of advice to NASA are dealt with in the next chapter.

First and Second Polar Years. Shortening the interval to 25 years would put the proposed Third Polar Year in the period 1957-1958, which would afford the added advantage of being a time of maximum sunspots, in contrast to the sunspot minimum of the Second Polar Year.

Among the new technologies that could be applied to the detailed investigation of the earth and sun were those of rocketry, whose applications to geophysics had become patently clear from the work of the Upper Atmosphere Rocket Research Panel, some of whose meetings Kaplan had attended. When the panel proposed to conduct rocket soundings as part of the IGY program, the U.S. National Committee quickly approved. Steps were taken at once to include a sounding rocket segment in the U.S. program for the IGY.

The National Research Council established a Technical Panel on Rocketry as part of the IGY machinery, and the National Academy of Sciences informed the CSAGI of the U.S. intention to use sounding rockets for geophysical investigations during the IGY.[5] By the time of the IGY planning meeting conducted by CSAGI in Rome during the week of 30 September 1954, rocket soundings had become an important element of the program.

But plans soon went beyond sounding rockets. When the U.S. sounding rocket program had begun in 1946, satellites were still deemed impracticable; now matters were different. The Navy, the Air Force, and other groups had continued to study the design and use of artificial satellites launched into orbit by large, powerful rockets, and by the early 1950s the feeling had developed that the satellite's time had come. Hidden by security wraps, some studies had moved fairly far along in the planning stage.[6] Von Braun and his people had convinced themselves that they could succeed in short order in orbiting a small satellite, and it rankled that official approval could not be obtained.

Members of the Upper Atmosphere Rocket Research Panel were aware of these studies, but those who were employees of the military did not feel free to press the issue. As has been seen, the panel recommended only a sounding rocket program to the Academy of Sciences. But geophysicist S. Fred Singer of the Applied Physics Laboratory, who had been conducting cosmic ray and magnetic field research in sounding rockets, felt under no restraints of military security. From some fairly simple calculations Singer concluded that it should be possible to place a modest (45-kilogram) satellite in orbit around the earth, and at every opportunity he urged that the country undertake to do so. Singer's conclusions were qualitatively correct, but his outspokenness generated some friction for at least two reasons. First, Singer's manner gave the impression that the idea for such a satellite was original with him, whereas behind the scenes many had already had the idea, and they felt that Singer had to be aware of this. Muzzled by classification restrictions, they could not engage Singer in debate. Second, being unable to speak out, those who had dug into the subject in much

greater depth could not point out that Singer's estimates overshot the mark somewhat, and that his suggested approach was not as workable as others that couldn't be mentioned.

Nevertheless, Singer did great service to those he made so unhappy. By making known the present possibility of placing artificial satellites in orbit, Singer aroused interest in this kind of device for scientific research.[7] The IGY was to be the first beneficiary.

Singer gained international attention for his proposal when, in August 1953 at the Fourth International Congress on Astronautics in Zürich, he described his idea for a Minimum Orbital Unmanned Satellite Experiment—which he called Mouse. Mouse would weigh 45 kilograms and would be instrumented for scientific research.

The International Scientific Radio Union, at its 11th General Assembly in the Hague, gave its support to Singer's proposal. At the urging of both Singer and Lloyd Berkner, on 2 September 1954 the Radio Union adopted a resolution drawing attention to the value of instrumented earth satellites for solar and geophysical observations. Later that month, on 20 September, the International Union of Geodesy and Geophysics at its 10th General Assembly in Rome adopted an even stronger resolution, actually recommending that consideration be given to the use of small scientific satellites for geophysical research.[8] Both the resolution of the Union of Geodesy and Geophysics and the earlier one of the Radio Union were conveyed to CSAGI, which held its third general planning meeting in Rome shortly after the close of the Geodesy and Geophysics Union meeting. Indeed, it is unlikely that these two resolutions could have been missed by CSAGI, since many persons attended all three meetings—radio, geophysics, and CSAGI—and even more attended both the last two. Also, the CSAGI membership included representatives from a number of scientific unions, including radio and geodesy and geophysics. The combining of forces to promote programs of mutual interest is traditional among the scientific unions, where maneuvering has much in common with ordinary politics.

At any rate, on 4 October 1954 CSAGI agreed and issued its challenge to the IGY participants in the following words, which closely parallel the resolution adopted by the Union of Geodesy and Geophysics:

> In view of the great importance of observations, during extended periods of time, of extra-terrestrial radiations and geophysical phenomena in the upper atmosphere, and in view of the advanced state of present rocket techniques, CSAGI recommends that thought be given to the launching of small satellite vehicles, to their scientific instrumentation, and to the new problems associated with satellite experiments, such as power supply, telemetering, and orientation of the vehicle.[9]

5

The Academy of Sciences Stakes a Claim

In the fall of 1957 the National Academy of Sciences in Washington was hosting an international conference on rockets and satellites. The mood was one of anticipation. The International Geophysical Year had begun officially on 1 July 1957 after several years of careful planning under the guidance of the Comité Speciale de l'Année Géophysique Internationale (CSAGI). Now, during the week of 30 September to 5 October 1957, CSAGI was giving special attention to the continued planning of the rocket and satellite part of the IGY program.

The International Geophysical Year—IGY for short—grew out of a suggestion made in 1950 by Lloyd V. Berkner to a small group gathered at the home of James A. Van Allen in Silver Spring, Maryland, that in the period 1957–1958 there should be a Third International Polar Year. Two previous International Polar Years, the first 1882–1883 and the second 1932–1933, had demonstrated in some measure the value of international cooperation in earth science investigations.[1] The group was heartily in favor of the idea.

No better promoter for such a project could have been found than Lloyd Berkner. A world-renowned geophysicist, he had long worked with problems of the ionosphere, the electrified region of the upper atmosphere that is responsible for reflecting radio signals around the world, making radio communications beyond the horizon possible. Berkner had been associated with G. Breit and M. A. Tuve who, at the Carnegie Department of Terrestrial Magnetism in Washington, had been among the first to measure the height of the ionosphere.[2] Berkner was interested in the institutional and international aspects of science, serving as adviser to the Department of Defense and in the State Department, and becoming very active in a number of the unions of the International Council of Scientific Unions. Most notable was his boldness of vision, which in scientific circles was fully a match for that of von Braun in rocketry.

Immediately swinging into action, Berkner and Sydney Chapman conveyed the proposal to the Joint Commission on the Ionosphere of the International Scientific Radio Union, the International Union of Geodesy and Geophysics, and the International Astronomical Union. In time the recom-

mendation reached and was adopted by the International Council of Scientific Unions—parent body of the various individual unions—which in 1952 appointed a special committee to oversee the project. In the course of soliciting participation, the enterprise was enlarged to encompass the scientific study of the whole earth, a subject more broadly appealing than polar investigations.

Thus the IGY was born.[3] The special committee was formally designated the Special Committee for the IGY, referred to in both speech and writing as CSAGI (generally pronounced *kuh sah jee*), the acronym of its name in French. Chapman became the president of CSAGI, Berkner its vice president, and both labored tirelessly and effectively to make the project go. Eventually 67 countries joined.

Chapman gave some idea of the scope of the IGY as finally conceived by its planners and organizers in his general foreword to the first volume of the *Annals of the International Geophysical Year*:

> The main aim is to learn more about the fluid envelope of our planet—the atmosphere and oceans—over all the earth and at all heights and depths. The atmosphere, especially at its upper levels, is much affected by disturbances on the Sun; hence this also will be observed more closely and continuously than hitherto. Weather, the ionosphere, the earth's magnetism, the polar lights, cosmic rays, glaciers all over the world, the size and form of the earth, natural and man-made radioactivity in the air and the seas, earthquake waves in remote places, will be among the subjects studied. These researches demand widespread *simultaneous* observation.[4]

Responding to the international call to countries to join in the IGY project, the National Academy of Sciences, through its National Research Council, established a National Committee for the IGY and selected Joseph Kaplan as its chairman (app. E). Kaplan, a geophysicist who had acquired a considerable reputation working in the laboratory on band emissions from various atmospheric gases, was noted for an inexhaustible supply of pleasant anecdotes. His genial personality was ideally suited to working with the difficult, dark, moody, sometimes abrasive Hugh Odishaw, executive director of the committee. Odishaw was the guiding genius behind the organization of the U.S. IGY program, and his influence could be felt in the international arena as well. He had a remarkable ability to foresee the future consequences of present actions, and was invaluable in mapping out strategy and directing tactics for securing support from the administration and Congress and in dealing with the inevitable conflicts and politicking on the international scene.

In putting forth his original proposal, Berkner had cited the great advances in technology and scientific instrumentation since the early 1930s—much of it generated in the prosecution of World War II—as a compelling reason for not waiting out the 50 years that had intervened between the

It remained for interested countries to respond to the recommendation. The United States had already announced its intention to conduct sounding rocket flights as part of the IGY program, but the complexity and expense of an earth-satellite program needed careful consideration by the agencies that would be expected to carry out the necessary development and IGY operations. Moreover, at the opening session of the Assembly of the Union of Geodesy and Geophysics Sydney Chapman as president of CSAGI had found it necessary to point out that the Soviets had not yet seen fit to join the IGY program. It was the United States and the Soviet Union, of course, that were expected to respond positively to the CSAGI proposal.

The U.S. National Committee for the IGY gave careful consideration to the proposal during the spring of 1955. Support was not immediately unanimous. Clearly the dimensions of this undertaking would be of a different order from the sounding rockets already a part of the IGY planning. Doubts were expressed over the wisdom of including the project in the IGY. Technical aspects were not the only considerations. There was also the concern about what would be the reaction of people to the launching of an artificial satellite that could easily be viewed as an eye in the sky, could well be accorded some sinister import, perhaps even be equated with some kind of witchcraft. Memories of Orson Welles's Mars invasion had by no means vanished. Most, however, favored endorsing the project. Joseph Kaplan, chairman of the committee, was especially enthusiastic and jokingly coined the phrase "Long Playing Rocket" for the satellite, by analogy with the long-playing records newly on the market. He suggested that, since sounding rockets had become familiar, the idea of a long-playing rocket would prove less disturbing than the completely new concept of an artificial satellite.

After much thought the National Academy of Sciences, sponsor of the U.S. IGY program, and the National Science Foundation, which provided the money, agreed jointly to seek approval of an IGY scientific earth-satellite program. The two agencies were successful, and the U.S. intent to launch an earth satellite during the IGY was announced from the White House on 29 July 1955.[10] A significant factor in administration support was the perceived need to develop and explore satellite capabilities for possible military applications. The Pentagon was assigned the job and, after a review of several possibilities, selected the Navy's Vanguard for the purpose.[11]

With scientific satellites now in the IGY program, the IGY committee established a Technical Panel on the Earth Satellite Program, to do for satellites what the rocketry panel was doing for sounding rockets. Richard Porter, the General Electric Company engineer in charge of the V-2 test program at White Sands, was asked to be chairman.[12]

When the Fourth Assembly of CSAGI met in Barcelona, 10-15 September 1956, the Soviet Union had joined the IGY and was prepared to say something about Soviet rocket and satellite plans. On 11 September, Prof. I. Bardin, speaking in Russian, announced to the CSAGI delegates that the USSR would have a rocket program in the IGY, details to be given later, and also would use satellites for pressure, temperature, cosmic ray, micrometeor, and solar radiation measurements.[13] Whereas the United States undertook to describe in considerable detail its sounding rocket and earth satellite plans, to aid those who wished to make correlated measurements by other techniques, the Soviet Union furnished little advance information.

Thus, when CSAGI convened the conference on rockets and satellites in Washington in the fall of 1957, there had been considerable time for work on the rocket and satellite projects, but it remained to be seen how much cooperative research could be done in association with those projects.

The subdued sense of anticipation that pervaded the sessions stemmed from the awareness that preparations had been under way for some time, that the IGY was already in full swing, and that the first artificial satellite must soon appear over the horizon. But those expectations did not diminish the surprise and dismay felt by U.S. scientists when the launching of *Sputnik 1* was announced on the evening of 4 October 1957. At the time many of the conference attendees were guests at the Soviet Embassy. The news, which had been broadcast by Moscow Radio, was brought to Berkner, who shared it at once with those present. His announcement practically wiped out the party as the U.S. scientists, in particular, scattered to their home bases to take stock of what had happened. The author had gone home that evening from the planning sessions at the academy and was about to call it a day when Hugh Odishaw called. As executive director of the U.S. National Committee for IGY, Odishaw wondered if a few people shouldn't meet at the IGY headquarters—1145 19th Street, N.W.—to discuss the turn of events. Once there, Odishaw, Richard Porter, the author, and others kept track of the Soviet satellite's course. From radio sightings as they were reported, the ground track of Sputnik was plotted on a map in the office, and in a few hours a pretty good idea emerged of the kind of orbit Sputnik was following.

As the group in imagination followed the course of the satellite across the heavens, the members tried to weigh the Soviet accomplishment against the fact that the launching of the U.S. satellite, Vanguard, was still some months away. They tried to estimate what the public reaction would be. Disappointment was to be expected, but they did not anticipate the degree of anguish and sometimes genuine alarm that would be expressed over the weeks and months that followed.

The next morning, Saturday, 5 October 1957, in the auditorium of the U.S. National Academy of Sciences, Anatoly Blagonravov took the floor to speak at length about Sputnik. Understandable pride was evident in Blagonravov's bearing, but his words also bristled with barbs for his American listeners. The speaker could not—or at any rate did not—refrain from chiding the United States for talking so much about its satellite before having one in orbit, and commended to his listeners the Soviet approach of doing something first and then talking about it.

While there was some measure of justice in Blagonravov's ungracious comments, his U.S. colleagues couldn't help feeling that he missed—perhaps intentionally—the point that much of the advance discussion of the U.S. IGY satellite program was to provide necessary information for planning by those who wished to cooperate in the tracking or other operational aspects of the mission. In view of the fruitlessness of CSAGI's efforts to elicit any such accommodation from the Soviets, either at Barcelona in 1956 or at the meetings in Washington, the remarks of their Russian colleague were doubly frustrating.[14]

Nevertheless, admiration for the Soviet achievement was genuine and universal, and his colleagues could heartily applaud when Blagonravov declared that he hoped that "this first step" would "serve as an inspiration to scientists throughout the world to accelerate their efforts to explore and solve the mysteries and phenomena of nature remaining to be explored."[15]

Reaction in the United States was strong and widespread. It was clear, albeit intuitively to most, that a new dimension had been added to man's sphere of thought and action. Equally clearly, something had to be done about the fact that the United States had not been the first to put a satellite in orbit. One read and heard talk about Soviet technological supremacy, U.S. loss of leadership, the missile gap, and security and economic implications. In view of the impressively large weights of *Sputnik 1* (80 kg) and *2* (508 kg, 3 Nov. 1957), and the multiton launch vehicles that they implied, the 8½-kg payload of *Explorer 1* launched on 31 January 1958 did little to allay such concerns. President Eisenhower attempted to downplay the Soviet achievement, but couldn't carry it off.[16] Congress took the matter seriously, largely through apprehension over military implications, and began to crank up the machinery to respond to what was viewed as a crisis. On his part, Eisenhower created the post of science adviser to the president, elevated his Science Advisory Committee to White House level, and asked the committee to develop a national policy on space. The result was to be the National Aeronautics and Space Act of 1958.

By now atmospheric and space science had moved far beyond the narrow confines of the Rocket and Satellite Research Panel and had established a base from which the space science program could proceed following the creation of the National Aeronautics and Space Administration in the summer and fall of 1958. From the membership of its technical panels on rocketry and on the earth satellite program, the academy established a Space Science Board in June 1958, to advise the government in what prom-

ised to be a fast-growing and important field. Lloyd Berkner was named chairman of SSB (app. F).

Events of the next three-quarters of a year after the first Sputnik launching make a fascinating and educational story as Congress and the administration cooperated and wrestled with each other to hammer out a legislative response to the crisis.[17] A number of circumstances combined to give scientists the civilian agency and open space program they favored. How this came about will be dealt with in chapter 7. But before proceeding to that part of the narrative, it is appropriate to pause and take stock of the rich harvest of scientific knowledge that a decade of rocket sounding had already produced before artificial earth satellites took on an importance that commanded the attention of the president and the Congress.

7

Response to Sputnik: The Creation of NASA

How brightly the Red Star shone before all the world in October of 1957! Streaking across the skies, steadily beeping its mysterious radio message to those on the ground, Sputnik was a source of amazement and wonder to people around the globe, most of whom had had no inkling of what was about to happen. To one nation in particular the Russian star loomed as a threat and a challenge.

In the United States many were taken aback by the intensity of the reaction. Hysteria was the term used by some writers, although that was doubtless too strong a word. Concern and apprehension were better descriptions. Especially in the matter of possible military applications there was concern, and many judged it unthinkable that the United States should allow any other power to get into a position to deny America the benefits and protection that a space capability might afford. A strong and quick response was deemed essential.

Actually, as has been seen in chapters 3 to 5, the United States was not far behind. A full decade of pioneering work had brought into being a respectable stable of rockets and missiles and still more powerful ones were under development, some of them nearing completion. Tracking and telemetering stations were operating, and a number of missile test ranges were functioning. Sizable teams of persons with capabilities pertinent to space research and engineering were available in both government and industry. And more than 10 years of sounding rocket research combined with open publication of results had given the United States a definite edge over the USSR in space science, in spite of its priority in the satellite program. Without question the United States was competitive in space even as the country deplored its loss of leadership.

Leadership was the key word. To be competitive was not enough. In an age when technology was vital to national defense, essential for solving problems of food, transportation, and health, and important to the national economy, technological leadership was an invaluable national resource not to be relinquished without a struggle. It was technological leadership that would generate a favorable balance of trade for the United

States and afford strength in international negotiations. Moreover, the *appearance* of leadership—while in no way equivalent to genuine technological strength in importance—nevertheless had a strong bearing on the international benefits to be won. President Eisenhower was right when he asserted that the country's position in rockets and missiles was a strong one, which the launching of a small scientific satellite by Russia could not substantively weaken, but in the mood of the times people were not disposed to listen.

One heard of a race with Russia, a topic that would be debated often in the years to come. While many would deny the necessity to run a race—and some would even contend that no race existed—for most, competition with the Soviets was serious business. Even those in a position to appreciate the strength of the U.S. position did little to bring it out, most likely because they, too, were persuaded of the importance of recapturing leadership in space, especially in view of the military implications. National leaders were worried about the obvious great size and lifting capacity of the Soviet missiles. Also, insertion of a satellite into orbit proved that the USSR had mastered the final ingredient of a successful intercontinental ballistic missile, guidance. Moreover, all this had occurred while the U.S. Atlas missile was still under development, far from deployment.

During those formative months of late 1957 and the first half of 1958 the broad spectrum of forces impacting on space matters—at once synergistic and conflicting—began to become apparent. In the face of the Soviet challenge academic, industrial, and political forces merged in a common conviction that the country must put its space house in order. The mutually reinforcing effect of these disparate interests all pushing for a properly organized, unified national space program led eventually to the creation of the National Aeronautics and Space Administration; their continued cooperation during the ensuing years produced the broad spectrum of achievements in manned exploration, science, and applications in outer space with which the world has become familiar. But the individual motivations—political objectives, commercial goals, professional aspirations—and the differing philosophical backgrounds of the industrialist, academician, legislator, administrator, soldier, scientist, and engineer set up cross currents and conflicts of varying intensity that run through the early years of NASA's history.

During the months preceding the passage of the National Aeronautics and Space Act of 1958, space science and potential military applications were already established areas of space activity that contended to attract the various interest groups that had or might have a stake in the nation's future in space.[1] Industry gravitated toward the military, with which it already had a close and profitable association. Professional societies, the Rocket and Satellite Research Panel, the Academy of Sciences, and the President's Science Advisory Committee naturally pursued the scientific

side of the matter. It was the military implications of space, the bearing that future space developments might have upon national defense and security, that imparted a sense of urgency to the deliberations in the executive and legislative branches. Communications satellites, weather satellites, and earth observations from space for intelligence were all important to the military, the last named being of special significance in the Cold War. These considerations caught and held the attention of the legislators. But, though the military implications were deemed the primary concern of the country, circumstances elevated the scientific aspects to a position of considerable influence.

A majority of those who would finally make the decision soon became convinced that the most effective way of proving U.S. leadership in space would be to demonstrate it openly.[2] Moreover, a space program conducted under wraps of military secrecy would very likely be viewed by other nations as a sinister thing, a potential threat to the peace of the world. A cardinal point in the U.S. military posture had always been that the development and maintenance of U.S. military strength was peaceful, not intended for aggression, but for self-defense and to enable the country to help maintain stability in a world in which weakness too often provided the occasion for trouble. It was an important thesis for the U.S. public to continue to believe and to sell to the rest of the world and, in a matter as portentous as space seemed to be, special efforts were needed to present the proper image. It seemed important, therefore, that the U.S. space program be open, unclassified, visibly peaceful, and conducted so as to benefit, not harm, the peoples of the world.

A logical conclusion of this reasoning was that the program should be set up under civilian auspices. Thus, although the military had by far the greatest amount of experience pertinent to conducting a space program, it was by no means a foregone conclusion that the Pentagon would be assigned the principal responsibility. To be sure, the Army, Navy, and Air Force had been among the earliest to study the usefulness of space to support their missions.[3] Military hardware afforded the only existing U.S. capabilities for space operations.[4] Moreover, the services had provided the funding and much of the manpower for the rocket-sounding program of the Rocket and Satellite Research Panel, many of whose members were civilian employees in military research laboratories, as shown in appendix A. Yet so powerful was the conviction that the program must project an image of benevolence and beneficence that the otherwise overriding military factors were themselves outweighed.

Reinforcing these views were President Eisenhower's own convictions. Already distressed over the enormous power and unmanageability of what he later called the military-industrial complex, Eisenhower was not disposed to foster further growth by adding still another very large, very costly enterprise to the Pentagon's responsibilities. Moreover, at the time the Pentagon did not enjoy the best of relations with Capitol Hill. One heard talk of a "missile mess" and interservice rivalry in the Pentagon. Such concerns led, during the very period when the administration and Congress were deciding America's role in space, to the appointment of a new secretary of defense, the creation of the Advanced Research Projects Agency, and passage of the Defense Reorganization Act, which among other things set up the Office of the Director of Defense Research and Engineering. Such considerations, plus Eisenhower's not seeing in Sputnik the crisis for national defense that others considered it to be, predisposed him to favor a space program with a strong scientific component under civilian management.

The scientists were united in their desire to have a strong scientific component in the space program. The greatly expanded federal funding of science in the years following World War II had declined. Members of the President's Science Advisory Committee and James Killian, special assistant to the president for science and technology, saw in the space program an opportunity to renew national support of science. Under the circumstances Killian and PSAC had a considerable influence in the creation of NASA, pressing for a space program under civilian management with a strong scientific flavor.[5] In this they were supported by the Rocket and Satellite Research Panel; the National Academy of Sciences, where President Detlov Bronk took a personal interest and where the Space Science Board was set up; by the American Rocket Society; and by other groups of scientists who felt impelled to speak out on the issue.[6]

Against this background the debate on how precisely to respond to the Soviet challenge proceeded. A deluge of proposals descended upon various congressional committees. In the Department of Defense the administration had set up the Advanced Research Projects Agency, approved by Congress in February as a temporary holding operation.[7] But, as pointed out, there were cogent reasons for a space program under civilian auspices—in which case provision would also have to be made to meet the vitally important military needs. Among the civilian possibilities was the creation of a new agency—which some of the scientists had recommended—but to those who knew what was involved, that was a horrendous undertaking. Alternatively one could assign the responsibility to an existing agency, or build a new agency around an existing organization as nucleus. With the application of nuclear power to rocket propulsion in mind, the Atomic Energy Commission was interested in taking on the job, as both the commissioners and the Joint Committee on Atomic Energy on the Hill made plain to members of the Rocket and Satellite Research Panel when they called to enlist support for the creation of a National Space Establishment.[8] But, for a number of reasons, the choice finally fell on the National Advisory Committee for Aeronautics (NACA).

The NACA would not have been the choice of most scientists. As a highly ingrown activity, the agency did not enjoy a particularly great

esteem in scientific circles, being thought of more as an applied research activity serving primarily industry and the military. Members of the Rocket and Satellite Research Panel, in particular, were skeptical of the ability of an agency almost entirely oriented toward in-house research and with no experience in the management of large programs to take on all the research, development, and operational tasks of a space program that some members thought would soon entail $1 billion a year. For most of its life NACA had managed at most a few tens of millions of dollars a year. In fact, the annual budget had not exceeded $1 million before 1930 and had not passed the $10-million mark until World War II. It took the construction and operation of the large wind-tunnels of the 1950s to push the budget toward $100 million. Skepticism within the panel was not lessened by the cautious attitude NACA management had displayed through the years toward letting NACA people take part, even in a small way, in the sounding rocket program. Doubts about the choice of NACA were increased in the months following the launching of Sputnik by conversations between panel members and NACA's Director of Research Hugh Dryden and Chairman James Doolittle.

The views of the scientists probably carried little weight. More telling was the disenchantment with NACA on the part of its own clients, the Air Force and industry. The agency had started in 1915 as an advisory group, as its name implied, but became gun shy when its advice began to generate at least as many enemies as friends.[9] As a consequence the NACA soon turned away from advising and toward research. Even here it was necessary to keep from treading on the toes of either industry or the military, and as a consequence the agency gravitated toward aerodynamic and wind-tunnel research, in which both clients were happy to have help. Over the years the agency had acquired a reputation of caution and conservatism. This conservatism may have caused NACA to miss out on a number of important aeronautical advances, the most significant of which was jet propulsion, where Britain and Germany took the lead. At any rate, because of such missed opportunities, NACA in the 1950s no longer had the unqualified endorsement of the military and industry that it once had, and in the view of at least one historian might well have died had not the space program come along to revive it.[10] Under the circumstances the agency was available, and it was a case of assigning responsibility for the space program to an organization whose future was otherwise in doubt.

NACA pursuit of this opportunity was something less than sparkling. At the urging of younger members of the agency, Dryden and his staff developed a number of papers on the subject of space research. On 14 January 1958 the so-called "Dryden Plan" was made public.[11] The title, "A National Research Program for Space Technology"—rather than a name referring to the exploration and investigation of space—reflected the agency's characteristic caution and narrowness of outlook. The plan was a hodgepodge born of a desire to keep much of NACA's old way of life while embracing the interests of both military and civilian groups. Under the plan the national space program would be a cooperative effort among the Department of Defense, the NACA, the National Academy of Sciences, the National Science Foundation, and various private institutions and companies. The Department of Defense would be responsible for military development and operations, the National Academy of Sciences and the National Science Foundation would have responsibility for the scientific experiments to be conducted, mostly by the outside scientific community, while NACA would be responsible for research and scientific operations in space. This cautious approach—which courted everyone and satisfied no one—was endorsed in a resolution passed by the Main Committee of NACA on 16 January. On 10 February 1958 the agency issued an internal document giving details of the expansion of NACA that would be required to support the Dryden plan.[12]

In spite of the negative feelings about NACA, the availability of the agency, coupled with doubts about its future in the field of aeronautics and the desire to put the space program in civilian hands, eventually made NACA the prime candidate for the job. On 5 March Chairman Nelson Rockefeller of the President's Advisory Committee on Government Organization, Director Percival Brundage of the Bureau of the Budget, and Special Assistant for Science and Technology James Killian jointly delivered to President Eisenhower a memorandum recommending that "leadership of the civil space effort be lodged in a strengthened and redesignated National Advisory Committee for Aeronautics." The memo listed a number of liabilities, but stated that these could be overcome by enacting appropriate legislation. The NACA would be renamed the National Aeronautical and Space Agency, and the 17-member governing committee—which NACA insisted was the kind of buffer a research agency needed at the top to shield it from external forces—would remain, but the membership would be changed and its power reduced.

That same day President Eisenhower decided to build "a civilian space agency upon the NACA structure."[13] From that point matters moved rapidly within the executive branch. The Bureau of the Budget prepared draft legislation with assistance from Killian's office and NACA. The pace with which this was accomplished left little time for coordination with other agencies such as the Department of Defense, a matter that aroused considerable criticism during the congressional hearings on the bill. On 2 April 1958, Eisenhower submitted his proposal to Congress. The Bureau of the Budget had insisted on a single responsible head for the new agency, one who would be advised by a board of experts but would not be responsible to and shielded by such a board. NACA leaders disagreed, and according to Arthur Levine some members of the agency sought help from friendly congressmen to preserve the traditional NACA organizational pattern.[14] But although the administration's bill was considerably tighter than the diffuse approach of the Dryden plan, and although the presentation of the bill served to channel the congressional deliberations into the course that led to the passage of the National Aeronautics and Space Act of 1958, both committee members and witnesses found much in it to criticize.

> The original bill lacked provisions dealing with Congressional oversight and control, international cooperation and control, patents, indemnification, limitation of liability, conflict of interest, definition of terms, ceilings on salaries, relations with the Atomic Energy Commission, formal liaison committees, and over-all policy determination and coordination. There was no provision for nuclear propulsion, or even any recognition of its importance in this new field. Vagueness regarding the delineation of military and civilian activities in outer space was charged by many. There was no formal provision for determining agency jurisdictions in space research or settling of jurisdictional disputes. There was much criticism of the lack of clarity in the size and makeup of the board proposed in the Administration bill. Concern was voiced over the lack of substantive provisions backing up various aims put forth in the declaration of policy.[15]

In the end Congress adopted a bill which, while it accepted much of what the administration had proposed, nevertheless introduced substantial changes to meet the various criticisms.

The remarkable congressional response to the Sputnik crisis has been analyzed by a number of authors.[16] Even before President Eisenhower showed any willingness to take the matter seriously, Congress had begun to probe the subject of the nation's missile and satellite programs.[17] The Preparedness Investigating Subcommittee of the Senate Committee on Armed Services opened hearings on 25 November 1957, continuing through 28 January 1958, accumulating more than 7000 pages of printed testimony largely devoted to how the United States and the Soviet Union compared in science and technology in general and rockets and missiles in particular. In his opening remarks the chairman, Lyndon B. Johnson, set a tone of bipartisan, nonpolitical searching for the best possible national response to the Russian challenge, a tone that was to characterize the entire process of the next half year leading to the passage of the NASA Act. The unanimous report from these first hearings called for quick and vigorous action. Indeed, the clear determination of the Congress to do something about the crisis had much to do with goading Eisenhower into action to develop an administration proposal.

At first congressional investigation and study, while extensive and much to the point, showed little agreement on how to proceed. Numerous resolutions and bills were offered, some of them proposing the establishment of a permanent space organization.[18] For a while there seemed to be too many cooks, but in February 1958 matters began to gel. Senate Resolution 256 on 6 February created a Special Committee on Space and Aeronautics to frame legislation for a national program of space exploration and development. On 10 February, 13 senators, comprising a powerful representation of the Senate leadership, were named to the committee.[19] The membership included the chairmen and ranking minority members of all major Senate committees concerned. On 20 February, in a rare break with tradition, the majority leader, Senator Johnson, was elected chairman.

The House soon followed suit and on 5 March established its own blue ribbon group, the Select Committee on Astronautics and Space Exploration, to which 13 members were appointed.[20] As in the Senate, the House regarded the matter as sufficiently important to set aside tradition, and Majority Leader John W. McCormack was named chairman. Minority Leader Joseph W. Martin, Jr., was picked as vice chairman.

Meanwhile the administration had been preparing the draft legislation. The appearance of the administration bill drew congressional activity into focus. On 14 April Senators Johnson and Bridges introduced the bill as S. 3609. The same day McCormack introduced it in the House as H.R. 118811, with identical bills being put forth by eight other representatives.[21] The House committee began hearings the next day, 15 April, and continued them through 12 May. Not having conducted a previous inquiry, as had the Senate, the House hearings were thorough and extensive. In contrast, the Senate committee directed its inquiry more narrowly at the proposed draft legislation. The Senate hearings covered six days, opening 6 May and closing 15 May.

Many complex issues were debated: the organization and salary structure of the new agency and its location in the executive branch; the matter of policy guidance at the top, and how to provide coordination and liaison between the civilian space agency and numerous other activities—like the Department of Defense and the military services, the Weather Bureau of the Department of Commerce, the National Science Foundation, the Atomic Energy Commission, and the Department of Health, Education, and Welfare, for example—which had legitimate and important interests in space research and applications; and the matter of ensuring the military the necessary freedom of action to pursue applications of space that were deemed of military significance.[22] The last-named issue was of great concern and brought in by implication the question of how to divide responsibility in the space program between a civilian agency and the military establishment. Numerous other issues also had to be ironed out, such as organization within Congress and how to provide for congressional oversight, policy on information and publicity, and how to handle international matters such as cooperation in space.[23] Both committees felt that the administration proposal failed to cover adequately many of the important issues. As a consequence, the bill finally passed differed considerably from that initially proposed.[24]

Most significant for space science, Congress did not prescribe the specific content of the space program with which the NASA Act was concerned.

> In the end the legislative formulation of a detailed space program was bypassed. The legislation set up an agency, created its machinery, and provided for coordination and cooperation between it and other branches of the Executive.[25]

The Congress had found it impossible to divide the program between the military and the new civilian agency:

> It rapidly became evident that it was the use made of it and not the satellite itself which might well determine whether it would be of a military or a peaceful nature. For example, a reconnaissance satellite could be used to map and photograph the surface of the earth for purposes of defense or attack. It could also provide vastly improved means for the study and exploration of the universe.[26]

Intelligence gathering was generally conceded to be entirely military, space science essentially civilian—although the military would necessarily be interested in certain aspects of space science. All else was contested: manned spaceflight, launch vehicle development, and applications like communications and meteorological uses of satellites. In the face of this dilemma the legislators chose to provide a framework that would give both the military and the civilian space agencies the necessary freedom of action, while requiring coordination and mutual assistance. Having established the framework, Congress would leave it to the two agencies to work out between them the appropriate division of labor and responsibility—precluding, of course, unwarranted duplication of effort.

The lack of a specifically prescribed program gave the first administrator of NASA a wide degree of latitude in selecting projects and missions to undertake, a freedom of choice that was but little curtailed by guidance that James Killian supplied in the summer of 1958, assigning manned spaceflight, meteorology, passive communications, and science to NASA, and active communications and reconnaissance to the Department of Defense. The latitude NASA enjoyed permitted the development of a broad-ranging program of science and exploration, and the accompanying development of technology and the application of space techniques to practical uses. During the first several years this situation was entirely in keeping with the spirit of the times, and on the Hill there was more questioning of whether NASA was being bold enough than there was concern about overstepping any bounds. In fact, it was conservatism within the administration that led to considerable moderation in building up the program.

The climate was ideal for the growth of a space science program. Not being prescribed in detail—as far as science was concerned the NASA Act simply called for "the expansion of human knowledge of phenomena in the atmosphere and space"[27]—the science program could be permitted to unfold in keeping with the scientific process. Relying on the nation's scientists, including the National Academy of Sciences, NASA proceeded to attack the scientific problems of the atmosphere and space that the scientists themselves deemed most important and most likely to produce significant new information. The organization of the space science program, the establishment of advisory committees, the agency's funding requests, and the means by which individual scientists, universities, and other research organizations were invited to participate—all were designed to make the space science program a creature of working scientists, in the conviction that such an approach would produce the best possible program for the country.

In many ways, although it didn't always seem so to the scientists, space science occupied a favored position. As a means of diverting attention from the military overtones of the Sputnik crisis, President Eisenhower had favored a national space program with a scientific complexion. During the months of discussion on the Hill, there never arose the slightest question but that space science would be an essential element of the national space program. Long lists of scientists were called as witnesses, or their opinions sought by letter as to what to do. The importance of science to the program and the importance of a civilian arena for science, plus the international character of science, contributed to the argument for placing the space program in the hands of a civilian agency. Reinforcing such considerations in the minds of congressmen and senators was the image of success science had acquired in the International Geophysical Year that had brought forth the Sputnik challenge.

Of course, the freedom that the first administrator of NASA enjoyed in developing the civilian space program had also been accorded the military services in pursuing military interests in space. As already mentioned, it was the military potential of space that aroused the concern and held the attention of many legislators, and that virtually guaranteed a formally designated national space program. But the broad overlap of common interests that had stymied the legislators in their efforts to effect a satisfactory division between the civilian and the military in the first place was a potential source of conflict between the new agency and the military services. Such conflict the National Aeronautics and Space Council and especially the Civilian-Military Liaison Committee, called for in the NASA Act, were intended to handle.

Another feature of the NASA Act that was of importance to space science was the provision of a single responsible head for the agency. Under the pressure of a national clamor to close the gap with the Russians in space—a pressure continually reinforced by the urging of Congress to get on with the task—NASA had its best chance to break away from the con-

servatism that had characterized its predecessor. To continue the old NACA structure, as NACA officials had urged, with an advisory board determining policy and shielding the director from many of these outside pressures, might well have had a greater impact on science than on other aspects of the space program. Boards and committees tend to be conservative. Paradoxically, scientists as a class are quite conservative. As a group they would doubtless have been content to move more slowly, more cautiously, less expensively, making the most of the tools already developed in preference to the creation of larger, more versatile—and more expensive—tools. Exposed directly to the outside pressures to match or surpass the Soviet achievements in space, NASA moved more rapidly with the development of observatory-class satellites and the larger deep-space probes than the scientists would have required (chap. 12). Some of the most intense conflicts between NASA and the scientific community arose later over the issue of the small and less costly projects versus the large and expensive ones—a conflict that NASA's vigorous development of manned spaceflight exacerbated.

Of course, the scientific community is not monolithic, and there were so many widely differing opinions on these matters as to make speaking of a single position of the scientific community nonsense. Nevertheless it seems clear that the new organizational structure prescribed for NASA not only helped NACA people drop much of their conservatism, but also had an impact on the space science program in effecting a faster development of more advanced space tools than many leading scientists would have called for.

The National Aeronautics and Space Act of 1958 was a remarkable piece of legislation, and the process which produced it even more noteworthy. The thoroughness with which the subjects of space and its potentials and implications were investigated and studied, the thoughtfulness given to the issues raised, and the care taken in responding to the crisis precipitated by Sputnik provide a model that could well be commended as a pattern for the handling of legislative matters. As a practical matter, however, it is not likely that the Congress could find the time and resources to devote such attention to more than a select few of the issues that come before it. Also, few other issues are so free of partisan concerns and vested interests.

At any rate, the act provided an effective framework for both the civilian and military components of the nation's space research and exploration. In the course of time, some changes were found desirable.[28] Perhaps the most telling were those in coordination, the area in which Congress had displayed so much concern and on which so much time had been spent. President Eisenhower made little use of the Aeronautics and Space Council and did not provide a permanent staff for it, so it was left to NASA and the Bureau of the Budget to do the staff work. In April 1961 the NASA Act was amended to place the National Aeronautics and Space

Council in the Executive Office of the President, to replace the president with the vice president as chairman, to decrease the size of the council, and to broaden its functions to include cooperation "among all departments and agencies of the United States engaged in aeronautical and space activities."[29] Also the Civilian-Military Liaison Committee proved ineffective from the start. In September 1960 NASA and the Department of Defense jointly established an Aeronautics and Astronautics Coordinating Board, cochaired by the deputy administrator of NASA and the Defense Department's director of defense research and engineering. Because it worked, the AACB rapidly took over the functions of the Civilian-Military Liaison Committee. The new board succeeded because its cochairmen and members were in positions of authority in their respective agencies, where they could personally put into effect agreements arrived at in the board. No longer of any use, the liaison committee was abolished by reorganization in July 1965.[30] There were some other changes, and additional authorities were acquired from related legislation—such as the authority to award grants in support of basic science.[31] But, all in all, the strength and effectiveness of the NASA legislation lay in the original act of 1958.

Under its provisions NACA prepared to move out on its new career—as NASA. Dryden was not chosen as the first administrator. In retrospect it is easy to see why. The cautious and diffuse approach of the NACA with which Dryden was identified, and Dryden's conservative views on the budget needed by the new agency, did not jibe with the legislators' sense of urgency in space matters.[32] Instead of Dryden, T. Keith Glennan—president of Case Institute of Technology, former head of the Navy's New London Underwater Sound Laboratories, and for two years a member of the Atomic Energy Commission—was chosen. In spite of the difficulties with Congress, Dryden had an undiminished reputation for technical and administrative competence which led Glennan to ask specifically for him as his deputy.

After a brief preparatory period, Glennan officially opened NASA's doors on 1 October 1958. Space science was one of the first of NASA's programs to flourish. Nevertheless it was not the Sputnik crisis that brought space science into being. What Sputnik did achieve was to break out much of the U.S. space program, including space science, from under the military wing where it had resided during the pioneering years. Had it not been for the shock generated by Sputnik, the American space program would probably have evolved into one largely devoted to military objectives—with space science as an adjunct. Under such circumstances, in spite of the commendably enlightened policies of the U.S. military establishment regarding support of basic research, the free play of the scientific process would have been difficult to maintain. Pressures would have been in the direction of supporting research with military applications and imposing security classification on some of the results. With the program in NASA, the scientific community was in a stronger position to impress its brand on American space science and to work openly with foreign colleagues when that seemed appropriate.

Yet it is of interest that the members of the Academy of Sciences and of the President's Science Advisory Committee who had worked so hard to push the space program in the direction of science and toward the civilian arena were not those who proceeded to carry out the space science program. As leaders of the scientific establishment, they continued to be beset by the problems of maintaining adequate appreciation and support for science in general; and as soon as the space program was launched they returned to these broader matters. Rather, it was those who had already been engaged in rocket and satellite work, especially those working on projects connected with the International Geophysical Year, who began to develop the nation's space science program. These individuals, with years of experience behind them in industry, on the Rocket and Satellite Research Panel, and in the IGY program, naturally had proprietary feelings about space research; and it was easy for them to regard the space science program as very much their own creation. But the academy, from its association with IGY, and PSAC from its role in laying the legislative foundation for NASA, also had certain proprietary feelings about the program. There arose accordingly a tension—constructive for the most part—between NASA managers and advisers in the academy and on PSAC. The issues of what the space science program should be, how it should be carried out, and who should make the decisions arose early and recurred continually throughout the 1960s and into the 1970s.

8

NASA Gets under Way

None of the traditional conservatism of the National Advisory Committee for Aeronautics was evident in the autumn of 1958 when the National Aeronautics and Space Administration got under way. Rather, the industry, care, and thoroughness that had earned for NACA the respect of Congress over the years could be sensed as the new agency geared up for the challenges ahead. A seemingly endless list of matters had to be taken care of in the first few months after NASA was formally opened by Administrator Glennan on 1 October 1958, and everyone had his hands full.

The agency showed no inclination to take its role in the nation's space program for granted. The debates during the previous year about the importance of the space program and the country's poor position relative to the Soviet Union demonstrated that Congress would take a deep interest in what NASA did. Also, the significance of the choice of a new man, T. Keith Glennan, as the first administrator, rather than Hugh Dryden, the director of NACA, was not lost upon former NACA employees. Even though the National Aeronautics and Space Act of 1958 had given NASA extensive authority, the agency still felt the need to sell itself. As the staff prepared for NASA's first budget hearings, Abe Silverstein, director of spaceflight programs, admonished his people with words like the following: "Remember, it is not the program we have to sell. That has already been bought. What we have to prove is that we are the right ones to do it!"[1]

That was the mood of NASA as it bent to the tasks ahead. If anything stood out at the time, it was that everything seemed to be happening at once. In the white hot light of public interest, NASA had to establish its organization, expand its staff, acquire new facilities, find contractors for the work to be done, carry out Vanguard and the projects transferred from the Advanced Research Projects Agency, work out its relations with the military and other agencies, develop a budget, prepare for the first congressional hearings, and plan for the future—all while attempting to get a program immediately under way. Again it was Silverstein who put it into words: "Two years. It will take two years to get things really under control. After that you can begin to take it easy." As a prophet, Silverstein was half

right. It did take about two years to set NASA on the course it would follow for the next decade.

The jumbled character of NASA's first years is readily apparent in Robert L. Rosholt's review of the period;[2] but in the midst of all the scramble, things were getting done. From hour to hour, and from day to day, NASA managers would move from topic to topic, keeping things moving on all fronts. Gradually the program began to take shape. Space science, even though it had the advantage of a head start from the previous sounding rocket work and the scientific satellite program of the International Geophysical Year, shared in the growing pains of the new agency. In addition, problems peculiar to a scientific endeavor had to be solved.

The following pages take up a number of subjects that the National Aeronautics and Space Administration had to address itself to for all its programs, but here they are considered in the light of their bearing on space science. Although discussed under several topical headings, these matters were inextricably interwoven and were being worked on simultaneously.

Organization

President Eisenhower's decision of 5 March 1958 to build a civilian space agency around NACA set in motion the train of events that led to the establishment of NASA. On 2 April, when the administration's draft legislation was sent to the Hill, the president instructed NACA and the Department of Defense to work out the necessary plans. For its part NACA set up an Ad Hoc Committee on NASA Organization, under Ira Abbott, NACA assistant director for aerodynamic research, which made a preliminary report in May.[3]

The committee's suggested organization showed four major divisions: Aeronautical and Space Research, Space Flight Programs, Space Science, and Management.[4] The last named stemmed from a recognition that the prospective program would require substantial management attention, requiring, among other things, contracting for development and operations as well as for research. Aeronautical and Space Research would cover the advanced research of the NACA plus that pertinent to the investigation and exploration of space. The large development projects and operations required for the space program would be handled by Space Flight Programs.

Space Science remained a separate box on the organization chart through the tentative plan of 11 August 1958. In keeping with the plan that Dryden had proposed in January,[5] it was specially noted on the charts that the space sciences program would use the services of the scientific community, including the National Science Foundation and the National Academy of Sciences. On 19 August, Administrator Glennan met with key

NACA officials to go over the planning, and a provisional organization chart was issued on 21 August 1958, from which the space science box had disappeared. About this time the author began negotiations with Abe Silverstein for a number of the space scientists at the Naval Research Laboratory to join NASA. As an outcome of these negotiations, John Townsend, John Clark, and the author transferred to NASA Headquarters on 20 October 1958. A few days later, on 24 October, a tentative organization chart again showed a box for space sciences, but this time in the Office of Space Flight Development, under Silverstein. Glennan's first official organization plan in January 1959 retained space science in the Office of Space Flight Development.[6]

According to Glennan, one should not read too much into the shifting position and status of space science, which simply reflected the fact that "'space sciences' was only one of many organizational elements to be fitted together." Moreover, the administrator looked to Deputy Administrator Hugh Dryden to ensure that science was "accorded its appropriate role and status in the NASA family."[7]

Had the 21 August chart persisted, it is safe to say that the scientific community would have been most distressed. As it was, making space science a subsidiary of spaceflight development did not sit too well with key scientists, who did not hesitate to characterize science as one of the major purposes of the space program. At the 18 December 1959 meeting of the Space Science Panel of the President's Science Advisory Committee, for example, Chairman Purcell closed by declaring that "space science was the backbone of the American space program, the foundation of what we can do in applications."[8] Space science may have been put where it was in the fall of 1958 because the scientists from the Naval Research Laboratory and elsewhere who came into NASA were unknown quantities to Dryden and Silverstein.

Space science again disappeared from the organizational nomenclature when in February 1960 the author was listed as deputy to Silverstein. In its place were two titles: Satellites and Sounding Rocket Programs, and Lunar and Planetary Programs. Almost two years later, in November of 1961, the second administrator, James E. Webb, announced his first major reorganization of NASA; at that time the author became director of a newly created Office of Space Sciences, giving science the kind of visibility in the NASA organization that the scientific community felt it should have.[9]

An often repeated statement of NACA people was that the strength of NACA lay in its centers.* That was where the trained people, who represented the research and technical competence of the agency, lived. The same would be true of NASA. But from the outset Hugh Dryden was especially

*Langley Aeronautical Laboratory in Virginia, Ames Aeronautical Laboratory and the High-Speed Flight Station in California, and Lewis Flight Propulsion Laboratory in Ohio.

concerned that the research character of the NACA centers—on which NACA's reputation in aeronautical and aerodynamic research had rested—be preserved and protected against encroachment by the development and operational demands of the space program. Thus, the Office for Space Flight Programs had the dual purpose of providing new capability for space research and development, while leaving the old centers free to pursue the advanced research and technology that were their forte. This policy, which appeared in the earliest planning, persisted throughout the evolution of NASA, but weakened with the passage of time. Thus, to avoid overloading the Jet Propulsion Laboratory (which was already struggling with the Ranger, Surveyor, and Mariner projects), in the summer of 1963 management of the Lunar Orbiter was assigned to the Langley Research Center.[10] This sizable project was followed in the latter half of the decade by the even more demanding Viking.[11] When the Centaur rocket stage needed special attention to pull it through its development difficulties, the project was assigned to the Lewis Research Center.[12] At the Ames Research Center, studies of an astronomical satellite undertaken in 1958 and 1959 became the basis for much of the planning for NASA's Orbiting Astronomical Observatory.[13] Later Ames became the management center for the Pioneer projects.[14] The urge to take part in the space portion of NASA's program, the need for additional support to important projects, plus the argument that a modest development work would provide insights into technological needs that would benefit advanced research, militated against keeping the research centers "pure." It eventually became a matter of keeping the development work at a modest level.

Given the policy of protecting the research character of Langley, Lewis, and Ames, an entirely new capability for the unmanned and manned space programs had to be built. On the 29 January 1959 organization chart, Glennan listed under the Office of Space Flight Development, as space project centers: Jet Propulsion Laboratory, Beltsville Space Center, Wallops Station, and Cape Canaveral.[15] Wallops Station had been an arm of the Langley Research Center and would now be devoted to a variety of test projects, including the launching of sounding rockets and the Scout satellite-launching vehicle.[16] Cape Canaveral, of course, was the site of the Air Force's East Coast missile launching facilities, which would be expected to support NASA, as well as military, programs. The Jet Propulsion Laboratory was transferred from the Army to NASA by executive order on 3 December 1958, giving NASA a substantial capability for spaceflight development.[17] By mutual agreement JPL was steered in the direction of lunar and planetary exploration. The Beltsville Center—which took its temporary name from its location on surplus government land near the Beltsville Agricultural Center in Maryland—grew out of planning that had started before NASA was activated. This center was to provide a satellite research and development arm for the agency.[18] On the first of May 1959, just a week after construction had begun, Glennan announced that the new center would be called the Goddard Space Flight Center in honor of Robert H. Goddard. By September the first building was fully occupied. The center was dedicated on 16 March 1961. Goddard, the Jet Propulsion Laboratory, and Wallops Island were to become the principal NASA centers in the space science program, although as mentioned earlier, other centers contributed substantially.

STAFFING

NASA's task of assembling a team for the space program was helped immeasurably by being able to build on the NACA team. Abe Silverstein, brought to Washington by Hugh Dryden, played a key role in the pre-NASA planning and in getting the space program under way. His imprint was to be found on most aspects of the program, including space science.

Abe Silverstein was a hard-nosed, highly practical, boldly innovative engineer, with a solid conviction—consistent with NACA tradition—that all research had to have a firm justification in practical applications to which it would ultimately contribute. Abe had come from the position of associate director of the Lewis Propulsion Laboratory in Cleveland to NACA Headquarters to help plan the new NASA. He remained a few years to head NASA's Office of Space Flight Development and later the Office of Space Flight Programs, in which NASA's space science division was then located. With him from Lewis he brought a number of persons who were to play key roles in NASA's management structure: Edgar M. Cortright, for many years deputy in the Office of Space Science and Applications and later director of the Langley Research Center; DeMarquis D. Wyatt, a leading figure in programming and budgeting for the agency; and George M. Low, who took over the Apollo Project Office at the Manned Spacecraft Center after the tragic Apollo fire at Cape Kennedy. Still later, Low became deputy administrator of NASA. Abe also drew upon other centers in NACA, selecting Robert Gilruth of Langley to manage a manned flight Space Task Group, which evolved into the Manned Spacecraft Center, subsequently renamed the Johnson Space Center. From the Ames Research Center he chose Harry Goett to take over the directorship of the Goddard Space Flight Center after John Townsend had that enterprise well under way.

The space science team grew largely from researchers who flocked to NASA from other agencies. The author's upper-air-research colleagues at the Naval Research Laboratory comprised an appreciable number of these. Soon after transferring to NASA, John Townsend, who had been head of rocket sonde research at NRL, was given the task of bringing the Beltsville Space Center into being. Negotiating with the director of the Naval Research Laboratory, Townsend worked out the details of the transfer of additional scientists and engineers from his former branch, 46 of whom

were placed on the NASA rolls on 28 December 1959.[19] From NRL Townsend also secured temporary housing for the new NASA group. With the members of Project Vanguard who were transferred en masse by President Eisenhower on 1 October 1958, these employees accounted for most of the original staffing of the center. The manned flight Space Task Group at Langley was administratively assigned to the Goddard Space Flight Center for a while, but before any physical transfer took place the group was sent to Houston in 1961 as the nucleus of the Manned Spacecraft Center.

When Robert Jastrow, a physicist interested in properties of the upper atmosphere, transferred from the Naval Research Laboratory on 10 November 1958, he immediately set to work helping to plan the future space science program. An attractive, able scientist, Jastrow quickly earned the support of the administrator's office. He took the lead in developing for NASA a theoretical space sciences group, from which eventually came both the Theoretical Division and the Institute for Space Studies of the Goddard Center. Through both of these activities Jastrow was instrumental in drawing a high level of scientific talent into the agency, either onto NASA rolls or as visiting scientists.

Remaining at headquarters, John Clark and the author worked with Morton Stoller, Edgar Cortright, and other NACA people to build up a space sciences staff. Nancy Roman was enticed to leave the Naval Research Laboratory radio astronomy group to put her hand to developing an astronomy program for NASA. To help plan lunar and planetary programs, Gerhardt Schilling shifted over from the Academy of Sciences, where he had been associated with the International Geophysical Year and Space Science Board staffs. Robert Fellows, a chemist, came from Sprague Electric Company to join in planning and directing the upper-atmosphere research program.

Such was the pattern, but by no means the full accounting, of the early space science staffing of NASA. Those who had been pioneering in space research and development swelled the rolls of workers in the space program, both within and outside of NASA. And a great many of these were scientists interested in taking part in the space science program.

PROGRAM

The purpose of an organization and staff is to do something. Ideally one should know what that something is before trying to shape an organization or to hire people, for the program should determine how and with whom to go about it. In practice the ideal can hardly ever be attained. In NASA much of the planning had to be done as the agency organized itself and hired staff. But not entirely, because NACA people—and others—had thought a great deal about what should be included in the space program.

In anticipation of the use of near-earth satellites for geodesy, during the 1950s a number of groups had been busily engaged in preparatory

Table 1
Pre-Sputnik Ideas for Space Projects

Project	Advocates
Sounding rocket research	Robert Goddard. Wernher von Braun. Upper Atmosphere Rocket Research Panel. Russians.
Explorer-class satellites	Upper Atmosphere Rocket Research Panel. International Geophysical Year rocket and satellite groups.
Geodetic satellites	Dirk Brouwer. Luigi Jacchia. R. K. C. Johns. John O'Keefe. American Geophysical Union Committee on the Geodetic Applications of Artificial Satellites. Upper Atmosphere Rocket Research Panel.
Biosatellites	Heinz Haber.
Astronomical satellites (solar and stellar)	Lyman Spitzer. Fred Whipple. Upper Atmosphere Rocket Research Panel.
Weather satellites	Harry Wexler.
Communications satellites	Arthur Clarke.
Manned space stations and orbital bases	Konstantin Tsiolkovsky. Hermann Oberth. Wernher von Braun. Willy Ley. H. E. Ross.
Interplanetary manned space flight, including manned lunar missions	K. E. Tsiolkovsky. Wernher von Braun. A. V. Cleaver.

studies and analyses. Among them were John O'Keefe of the Army Map Service, Luigi Jacchia of Harvard, Dirk Brouwer of Yale, and their colleagues. For the International Geophysical Year the Smithsonian Astrophysical Observatory had taken the lead in developing and putting into effect plans to use data from the IGY satellite camera network. As an outgrowth of an informal committee organized at the suggestion of R. K. C. Johns, the American Geophysical Union Committee on the Geodetic Applications of Artificial Satellites kept geodesists informed of the possibilities of the new tools. A committee report issued in September 1958 shows that considerable thought had been given to the subject.[20]

As early as 21 November 1957 the National Advisory Committee for Aeronautics had voted to establish a Special Committee on Space Technology.[21] Chaired by H. Guyford Stever, dean of the Massachusetts Institute of Technology, the committee included several members from the Rocket and Satellite Research Panel, such as James A. Van Allen, its chairman; Wernher von Braun, leader of the German missile experts working

for the Army Ballistic Missile Agency in Huntsville, Alabama; and William Pickering, director of the Jet Propulsion Laboratory. The committee, assisted by a number of specialized subcommittees, formulated a space research program. Although the report, completed on 28 October 1958 after NASA was already operating, never was published, NACA people had had the benefit of the thinking of the committee and its various subgroups on a wide variety of subjects, including technology, space applications, the physical and life sciences, and manned spaceflight.[22]

With respect to manned spaceflight, both NACA and military agencies had been very active. NACA's prospective thinking in February of 1958 had envisioned the travel of man to the moon and nearby planets. During the summer of 1958 the Advanced Research Projects Agency was besieged with requests for support of Air Force and Army manned spaceflight proposals. But shortly after the passage of the NASA Act of 1958, President Eisenhower assigned the new agency the responsibility for manned spaceflight. In mid-September Administrator Glennan and Roy Johnson, head of the Advanced Research Projects Agency, agreed that the two agencies should work together on a man-in-space program, and to coordinate the activity they established a joint Manned Satellite Panel.[23] Robert Gilruth, who was to be a key figure in the Mercury, Gemini, and Apollo programs, was chairman. By virtue of this spade work, NASA was ready only one week after its opening to proceed formally with Project Mercury, the nation's first manned spaceflight mission.[24]

The roots of the space science program went at least as deeply into the past as did those of the manned spaceflight program. Taking over the Vanguard program, assuming responsibility for much of the nation's sounding rocket research, and acquiring the Pioneer deep-space probes from the Air Force, NASA had an ongoing space science program from its first day. Building on these activities and drawing upon their own experiences of the past decade, the rocket research scientists who had come into NASA were able to put together a plan that described in a general way what NASA would be doing in space science for the next two decades. The shape of the emerging program was evident in NASA's first hearings before Congress.[25]

To trace the evolution of the space science program in the thinking of the NASA planners is interesting. A sheaf of working papers in the NASA files for January 1959 gives an overall summary of the space science program as envisioned at that time.[26] The program was described in terms of the different scientific disciplines to which space research could contribute—for example: particles and fields, astronomy, atmospheres, and ionospheres. A more formal document, 10 February 1959, elaborated further, listing atmospheres, ionospheres, energetic particles, electric and magnetic fields, and astronomy as areas of research within the NASA space science program.[27] Within the larger categories was a detailed breakdown.

Atmospheric research had been a major part of the sounding rocket work of the past decade (chap. 6). It investigated the principal properties of the earth's atmosphere, such as the pressure, temperature, density, and composition of the atmosphere as a function of height and geographic location. Particularly important were the variations of these quantities with time, especially the variations caused by solar influences. With the prospect of sending spacecraft to other planets the comparative study of different planetary *atmospheres* would be immeasurably aided. Among the important solar influences to study were those causing atmospheric molecules to become ionized forming electrified regions known collectively as the ionosphere. Again, with the possibility of including other planets, the plural *ionospheres* was used. *Energetic particles* referred to planetary radiation belts, radiations in the interplanetary medium, and cosmic rays. *Electric and magnetic fields* in the upper atmosphere and space would be important aspects of relationships between the sun and the earth, and presumably other planets as well. There was considerable interest in studies of *gravitational fields* in connection with geodesy and the celestial mechanics of the solar system and particularly for investigation of various theories of relativity, for which it was agreed that satellites should be very effective. The areas of *astronomy* were the familiar ones of sun, moon, planets, and stars. With satellites measurements could be made in ultraviolet, x-ray, and other wavelengths that could not be observed at the ground.

Amplifying the general description of the program were several pages devoted to specific problems ripe for attack. For example, under atmospheres one asked what were the primary sources of energy affecting the high atmosphere, and what was the relationship of the Great Radiation Belt to the heating of the upper atmosphere. A question that was still not answered a decade later concerned the precise relations between the earth's surface meteorology and the upper atmosphere. Of particular importance was the problem of learning in detail exactly how the sun exerted its influence on the atmospheres of the earth and planets. For ionospheric studies it was regarded as important to obtain details on the structure of the highest regions. Of especial interest, one looked forward to the study of other planetary ionospheres. Typical among the problems under energetic particles and magnetic fields were those in pinning down exactly how solar particles behaved in the vicinity of planets and, in particular, how the aurora was generated. Under gravitational fields the first problem was to get a detailed analysis of the earth's field into its various components, since from this would flow the solution of many other problems. Likewise one looked forward to determining the gravity fields of other planets. In astronomy, the first problem was to learn the spectral emissions of the sun, stars, and interstellar medium. This selection of existing problems is only illustrative.

Even more detail could be found in a second document of February 1959 entitled "The United States National Space Sciences Program," which added biosciences to the list of space science areas.[28] Of particular interest was the indication that working groups would be established immediately in half a dozen program areas, others later as needed. The evolution of the idea of working groups as a means of drawing outside scientists into the program will be discussed in some detail in chapter 9.

A five-page paper of 30 March 1959 elaborated on plans for a biosciences program, including the establishment of working groups in the area. By mid-April a paper titled "National Space Sciences Program," plainly a derivative from the 4 February document, showed clearly the directions the program was taking.[29] Although this was only half a year after NASA began, the agency would follow those directions for the next decade.

This early thinking of NASA was reflected in the report the National Academy of Sciences made to the international Committee on Space Research at its second meeting held at The Hague, 12-14 March 1959. In this report the breakdown of NASA's February working papers was followed.[30]

Quite properly the planning began with a consideration of the scientific objectives to pursue, a listing of the important areas of research, and an assessment of the significant problems to attack. One can see this approach in the NASA documents of the first half of 1959. But a program—by which is meant a long term, continuing endeavor of rather broad general objectives—must be carried out in discrete steps. In NASA parlance such discrete steps were called projects. Thus, the astronomy *program*, which was to investigate the universe from above the atmosphere, might be expected to continue as long as space techniques could produce significant new information, and there was no foreseeable end to that. But a satellite *project* to measure gamma rays from the depths of the galaxy—which most certainly would further the astronomy program by shedding light on energetic processes in the galaxy—would be of limited duration, long enough to prepare, fly, and operate the satellite, and interpret the data obtained.

Even as NASA's program plans were being developed, the agency started numerous projects to conduct investigations ranging over the different areas of space science. In reporting on the program, then, it soon became possible to list ongoing projects in which the nation's scientists were participating. In April 1960 NASA's report on the space science program could go far beyond the generalities of the papers developed in early 1959.[31] The various parts of the program were integrated under a few broad objectives; namely, to learn more about sun-earth relationships, the origin of the universe and the solar system, and the origin of life. The specific disciplines supporting the broad objectives were given as aeronomy, ionospheres, energetic particles, magnetic fields, astronomy, gravitational fields, lunar sciences, planetology and interplanetary sciences, and micrometeorites and cosmic dust. The similarity to the earlier breakdown is evident, although a few new terms appear. Replacing atmospheres was *aeronomy,* a term coined by Sydney Chapman during the 1950s from the Greek words meaning "laws of the air." The term *planetology* had been introduced to mean the study of the body of a planet as distinguished from the investigation of its atmosphere and ionosphere. It was felt that dust in space might be an important factor in studying the origin and evolution of planetary systems and in designing spacecraft for interplanetary flight, hence the prominence given to it. Micrometeorites were simply cosmic dust particles that struck another body such as the earth or a spacecraft.

In the April 1960 paper a great deal of detail could be given on projects to support the various programs. Under each of the above disciplinary categories were listed physical parameters to be measured, instruments to be used, experimenters responsible, and sounding rockets, satellites, or space probes to be employed. For example, under aeronomy the 30-meter sphere of thin metallized plastic constructed by William O'Sullivan of Langley Research Center was listed as a sounding rocket experiment to measure upper-atmosphere densities. The sphere would be carried aloft, ejected at altitude, and inflated. Accurate measures of the air drag on the falling sphere would give the desired densities. O'Sullivan's sphere was the forerunner of the Echo passive-communications satellite launched on 1 August 1960, which observers around the world were able to follow with the naked eye.

Under energetic particles the document had a long list of experiments on radiation belts and the magnetosphere, some of them having already been carried out successfully. For the related area of magnetic fields, *Pioneer 5* was shown as having been launched into deep space in March 1960. Radio, gamma-ray, solar, and stellar astronomy projects were in the works, and a solar observatory satellite was actually under development. A considerable amount of work was indicated as under way in gravitational fields, relativity, and geodesy. A series of lunar probes was also listed.

From this point on, the space science program evolved pretty much along the lines already established. A continuing effort to keep a spark of life in the planning recast the program objectives in different words from time to time. For example, in March 1961 the principal areas in the space science program were grouped under the headings: the earth as a planet, the earth's atmosphere, solar activity and its influence on the earth, origin and history of the moon and planets, and the nature of the stars and galaxies.[32] By the mid-1960s, when NASA made a special effort in its authorization hearings to present the broad perspective of the space science program, the principal categories had been reduced to two: exploration of the solar system and investigation of the universe. Elaborating on these objectives

the author stated to the Senate Committee on Aeronautical and Space Sciences:

> The first category includes the investigation of our Earth and its atmosphere, the Moon and planets, and the interplanetary medium. The nature and behavior of the Sun and its influence on the solar system, especially on the Earth, are of prime importance. With the availability of space techniques, we are no longer limited in direct observations to a single body of the solar system, but may now send our instruments and even men to explore and investigate other objects in the solar system. The possibility of comparing the properties of the planets in detail adds greatly to the power of investigation of our own planet. Potentially far-reaching in its philosophical implications, is the search for life on other planets.
>
> The fundamental laws of the universe in which we live are the most important objects of scientific search. Space techniques furnish a most powerful means of probing the nature of the universe, by furnishing the opportunity to observe and measure from above the Earth's atmosphere in wavelengths that cannot penetrate to the ground. There is also the opportunity to perform experiments on the scale of the solar system using satellites and space probes to study relativity, to delve into the nature of gravitation, including the search for the existence of gravitational waves.[33]

However expressed, the basic substance of the program was remarkably stable.

In contrast to the overall program, one would expect the projects to change considerably as the years went by. But even here many projects had their origins in the thinking of the first few years. Table 2 lists the major space science, or science-related, projects up to mid-1968. For each project the dollar symbol indicates the first fiscal year in which money was specifically charged against the project, although money from supporting research or other general sources most likely had been spent earlier in exploratory work on the project. In some cases an asterisk is inserted to show how much earlier serious consideration of such a project had been under way. Of the 25 projects named, 22 (or 88 percent) were under way by mid-1962 in the sense that costs were being formally charged to them. More than three-quarters of the projects were begun or were being seriously considered during Glennan's tenure.

Considering the rapid development of the NASA program—on all fronts as well as in space science—and the wide range of projects, including launch vehicle development to be discussed later, that were set in motion during Glennan's term of office, it seems clear that the first administrator must be given the credit for setting NASA on the course that it followed for the next decade. Nevertheless, as he himself stated, Glennan was

Table 2
First Recorded Direct Obligations
to Space Science or Related Projects
(by fiscal year)

Project	1959	1960s						
		0	1	2	3	4	5	6
Sounding rockets	$							
Vanguard	$							
Explorers	$							
Physics and astronomy advanced research	$							
Lunar and planetary advanced research	$							
Bioscience advanced research			$					
Manned space science advanced research						$		
Orbiting solar observatories	*	$						
Advanced orbiting solar observatories				$				
Orbiting astronomical observatories		$						
Orbiting geophysical observatories		$						
International satellites		$						
Pioneer	$							
Ranger	$							
Surveyor			$					
Surveyor Orbiter						$		
Lunar Orbiter				$				
Mariner			$					
Voyager†			*				$	
Scout development	$							
Centaur development	$							
Delta development	$							
Mercury	$							
Gemini				$				
Apollo	$							

SOURCE: Jane Van Nimmen and Leonard C. Bruno with Robert L. Rosholt, *NASA Historical Data Book, 1958–1968*, vol. 1. *NASA Resources*, NASA SP-4012 (Washington, 1976), pp. 136–48. By the time a specific project appears as such in the financial records, generally a considerable amount of time (sometimes years) has been spent on advanced planning and research to lay the groundwork for the project.

* = already under consideration.
$ = first record of direct obligations.
† = never finished.

no "space cadet." His was just the right balance of conservatism and interest in space to make him congenial to President Eisenhower and acceptable to the Congress. "Thus, with strong support, when it was needed, from Eisenhower the Administrator's Office . . . with pushes—strong pushes—from Abe Silverstein and believable and solid advice from Newell and others—and strong pushes from the Congress—set the pace."[34] It may be said that Glennan set a strong but measured pace.

The effect can be discerned in the methodical way in which the space science program was made to unfold. NASA's first and natural step was to extend the sounding rocket work, and the Pioneer deep-space investigations already under way. The modest step from those to solar and astronomical observatories came next, although the Orbiting Astronomical Observatory proved to be a much bigger bite than the space science managers had imagined. The investigation of the moon would be much more demanding and costly than near-earth missions, and a serious commitment to a lunar science program came more slowly, even though Harold Urey, Nobel Laureate and renowned student of the moon and planets, had begun to press for such a program in the first few months of NASA's existence.[35] Also, to maintain what he considered the right pace, Glennan for a while showed a reluctance to discuss planetary missions except as plans for later, for the more distant future.

But "later" was not long in coming, as table 2 makes clear. Before Glennan left office NASA was engaged in space science projects that took in not only the earth and its environs, but also the moon and planets, the sun, and even the distant stars. One may surmise that Glennan, exposed to the pressures from both within and without the agency, and perhaps himself caught up in the enthusiasm of those around him, moved more rapidly than he had originally intended. At any rate, he turned over to his successor, James E. Webb, a well rounded program, well under way.

By the time Webb took office, the course of the space program for the next decade has been set. Even Apollo, under study since the start of NASA, had been commended to President Eisenhower in the last months of the Republican administration. Though Eisenhower did not approve, the ideas were there ready to be seized when President Kennedy came to feel that the successful accomplishment of an extremely challenging space mission would be important to U.S. prestige. The renewed sense of urgency that the Apollo decision bestowed on the space program made Webb's task one of loosening the shackles imposed by the previous administration and stepping up the pace. But the program content was already there. Thus, Apollo and Gemini may be looked upon as super projects designed to pursue an already existing program with greater vigor.

In this climate the space science managers put together plans to expand their program. On 22 May 1961 the Space Sciences Steering Committee, which consisted of NASA's principal space science program managers, met with selected consultants to review the proposed expansion.[36] The consultants represented a cross section of the disciplines of space science: Dirk Brouwer of Yale (astronomy), Joseph W. Chamberlain of Yerkes Observatory (atmospheric sciences and planetary astronomy), Robert A. Helliwell of Stanford (radio physics), Harry H. Hess of Princeton (geophysics and geology), Bruno B. Rossi of the Massachusetts Institute of Technology (high-energy physics and x-ray astronomy), and Harold C. Urey of the University of California at San Diego (lunar and planetary science). The group endorsed the expansion of the program proposed by NASA and emphasized a number of exciting researches to pursue, including the moon's gravity, the almost nonexistent lunar atmosphere, solar radiations, the vicinity of the sun as close as 16000000 km from the solar surface (one-tenth the distance from the sun to the earth), and micrometeoric particles in space. In characteristic fashion the scientists favored large numbers of small spacecraft for investigating the vicinity of the earth and heartily endorsed small grants from NASA to large numbers of universities for basic research. They recommended that serious consideration be given to a proposal from General Motors for obtaining by unmanned methods a sample of material from the moon.

By the following autumn NASA had moved forward substantially in the expansion of its program and was beginning to feel the need for a full-scale exchange with the scientific community on the content and course of the program. The Academy of Sciences was requested to organize a study of the space science program, which the Space Science Board agreed to do.[37] The study would be conducted during the summer of 1962; the program to be reviewed was described by the author at the NASA management conference held at the Lewis Research Center on 11 January 1962.[38] In the program were sounding rockets, satellites, and space probes. The Scout, Delta, Agena, and Centaur rockets, to be discussed in chapter 10, were included. Spacecraft, also to be taken up in chapter 10, included a variety of Explorers; solar, geophysical, and astronomical observatories; the lunar spacecraft Ranger and Surveyor; and Mariner planetary spacecraft. Some advance thinking about a spacecraft for a bioscence program was mentioned. The scientific fields were those already mentioned. Geodesy was described as important but stymied by difficulties over classification. An international cooperative program including many of the disciplines was well under way. What would be of special interest to the summer study participants was the university program, which in January of 1962 was rapidly increasing. The author's report listed university program funding as \$3 million for fiscal 1959, \$6 million for fiscal 1960, and \$14 million for fiscal 1961 (the fiscal year beginning July 1 preceding the corresponding calendar year). In fiscal 1962 plans were to use \$28 million on research projects in universities, largely flight experiments, plus \$12 million for support of graduate training in space related fields, research facilities on university

campuses, and grants for research of a more general nature than the specific flight projects. Space science managers projected a university program growing in the future to \$100 million a year in projects and about \$70 million a year in the broader grant program—a growth that was only about half realized in the 1960s.

In short order NASA's new team of leaders, which included many of the NACA's top people, remade the organization and activities acquired from the National Advisory Committee for Aeronautics into a National Aeronautics and Space Administration as called for in the NASA Act of 1958. As the new agency organized, developed its staff, and built its new facilities, NASA started the space science program along the lines that would be followed for the next decade. The rapidity with which this was done was both a tribute to the NASA team and convincing evidence that a strong base had existed on which to build in a number of areas, including space science. The thoroughness of the early work would be attested to by the fact that during the next decade—though the pace would be increased—little that was new would be added to the program.

long list of others. For some of these, interest in space science flared up at the very start, while for others the interest gradually emerged as the program unfolded.

Inheriting so much from the International Geophysical Year, NASA had an international program from the outset.[45] There were two main arenas, that of the international scientific circles such as the International Council of Scientific Unions and its newly formed Committee on Space Research, and that of a political nature, falling generally in the sphere of the United Nations. There were numerous political considerations relative to space, and NASA was immediately drawn into United Nations deliberations on space matters.

But the natural arena for space science was the international scientific community, and from the start NASA gave strong support to the Committee on Space Research. Among the unions of the council represented on COSPAR were the International Union of Scientific Radio and the International Union of Geodesy and Geophysics, which had first recommended the use of scientific satellites during the International Geophysical Year. Following the organizing meeting convened by the author in London in November of 1958, COSPAR held its first full-scale business session in The Hague, 12–14 March 1959.[46] At that meeting Richard Porter of the Space Science Board, U.S. representative to COSPAR, asked the author whether the United States might offer to launch space science experiments for COSPAR members. In a phone call to Washington, the author obtained Hugh Dryden's approval to inform the meeting that NASA would be willing to do so. Porter then wrote to President H. C. van de Hulst, saying that the United States would accept single experiments as part of larger payloads, or would launch complete payloads prepared by other countries.[47] The response to the U.S. invitation was immediate, and before the year was out a number of cooperative projects had begun. With the Soviet Union, genuine cooperation proved to be difficult during the 1960s, less difficult in the 1970s climate of detente. These subjects are discussed at length in chapter 18.

As the leaders of NASA worked to reshape the NACA into an aeronautics and space organization, they also laid the foundation for the many relationships with other government agencies, industry, and the scientific community that played an essential role in planning and conducting the program. But none of this would have been of any avail without the principal tools, the rockets and spacecraft essential to the investigation and exploration of space. A first order of business was to provide for these tools. That NASA set about to do, striving to overcome as soon as possible the visible gap that lay between the United States and the Soviet Union in propulsion capabilities and launchable spacecraft weights. Because of the central importance of launch vehicles and their payloads, the next chapter is devoted entirely to them.

10

Rockets and Spacecraft: Sine Qua Non of Space Science

Even as NASA was being formed, the stable of American sounding rockets was impressive. There were small (Deacon, Cajun, Arcon, Arcas), medium (Aerobee, Aerobee-Hi), and large (Viking) rockets. Viking had been designed to replace the V-2, which was no longer used after the test program ended in 1952. There were rockets using solid propellants (Deacon, Cajun, Arcon, Arcas) and rockets using liquid propellants (second stage of Aerobee, Viking). Multistage combinations (Nike-Deacon, Nike-Cajun, Aerobee) achieved higher altitudes than could economically be attained with single-stage rockets. Rockets had been launched from balloons, from aircraft, and from launchers aboard ships at sea. These sounding rockets and the high-altitude research program that went with them provided NASA with an immediately on-going component of its space science program.[1]

A similar situation existed with respect to the larger vehicles needed for launching spacecraft into orbit. The reentry test vehicle Jupiter C—which launched America's first satellite, *Explorer 1*, and which used the Redstone missile as its first stage—gave rise to a first group of what were called Juno space launch vehicles. Later versions of Juno used the more powerful Jupiter intermediate-range ballistic missile as the first stage.[2] The Redstone, which was created for the Army by the von Braun team and in which one could detect a distinct V-2 ancestry, was on hand and was used for America's first suborbital manned flights.[3] The Vanguard IGY launch vehicle, which used derivatives of the Viking and Aerobee sounding rockets as its first and second stages, was also available.[4] NASA took over Vanguard from the Naval Research Laboratory and completed the program, after which the Vanguard first stage was retired; but the upper stages were combined with the Air Force's Thor to create the Thor-Delta, or simply Delta, launch vehicle, which from the very start was one of the most useful of the medium-sized combinations.[5] In 1958 the Air Force's Atlas was the most powerful U.S. rocket that could be quickly pressed into service as a